Exact transport and packing dimension for the almost Mathieu operator*

Seth Douglas Nidhal Mghirbi

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Abstract

We determine the transport exponents and packing dimension of the almost Mathieu operator $(Hu)(n) = u(n+1) + u(n-1) + 2\lambda\cos 2\pi(\theta + n\alpha)\,u(n)$ for every $\lambda > 1$ and every irrational α. Let $\beta(\alpha) = \limsup_n q_n^{-1}\log q_{n+1}$ and $\gamma = \max\{0, 1 - \log\lambda/\beta(\alpha)\}$ ($\gamma = 0$ if $\beta(\alpha) = 0$, $\gamma = 1$ if $\beta(\alpha) = \infty$). For almost every phase, every moment of a packet started at one site has upper growth exponent γ, and so does its fastest polynomial tail. Transport thus switches on at the spectral transition $\lambda = e^{\beta}$ of Avila, You and Zhou. The lower bound holds at every phase. If $\beta(\alpha) > \log\lambda$ it is carried by a fixed fraction of the packet: for $c > h > \log\lambda$ and large n with $T = |q_n\alpha - p_n|^{-1} \geq e^{cq_n}$, a fraction at least $2^{-32}/(1 + 4\pi\lambda)$ lies beyond distance T^{1-h/b_n}, $b_n = q_n^{-1}\log T$; these exponents tend to γ as $b_n \to \beta(\alpha)$ and $h \downarrow \log\lambda$. For almost every phase, every nonzero spectral measure has upper local dimension γ almost everywhere, so both its packing dimensions equal γ; at every phase they are at least γ. For $0 < \log\lambda < \beta < \infty$ this makes the packing bound $2(1 - \log\lambda/\beta)$ of Jitomirskaya, Liu and Tcheremchantsev sharp with factor 1, for almost every phase. The proofs use an exactly conserved polynomial of H, an arcsine law for the rational discriminant, and long elliptic runs of the Weyl solution. As $\beta(\alpha)$ can be noncomputable for computable α, so can γ: at $\lambda = 2$, $\theta = 1/4$, the exponents realized by computable frequencies with eventually odd denominators are exactly the reals in $[0, 1]$ that are right-computably enumerable relative to the halting problem, and one explicit computable operator has exponent of Turing degree $\mathbf{0}''$.

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1 Introduction

The almost Mathieu operator

$$(H_{\lambda,\alpha,\theta}u)(n) = u(n+1) + u(n-1) + 2\lambda\cos\bigl(2\pi(\theta + n\alpha)\bigr)u(n) \tag{1}$$

on $\ell^2(\mathbb{Z})$ is the central model of quasi-periodic spectral theory. Physicists know it as the Aubry–André model [1], and it is realized with cold atoms in bichromatic optical lattices [21] and with light in quasi-periodic waveguide arrays [19]. Here $\lambda > 0$ is the coupling, α is an irrational frequency and $\theta \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$ is a phase. For $\lambda > 1$ the Lyapunov exponent on the spectrum equals $\log\lambda > 0$ [6], and the folklore physical picture is that every state is localized.

That picture depends on the arithmetic of α. Let p_n/q_n be the continued-fraction convergents of α and

$$\beta(\alpha) = \limsup_{n\to\infty} \frac{\log q_{n+1}}{q_n} \in [0, \infty]. \tag{2}$$

Avila, You and Zhou [2] proved that the supercritical regime splits sharply at $\lambda = e^{\beta(\alpha)}$. For $\lambda > e^{\beta}$ the operator has Anderson localization for almost every phase. For $1 < \lambda < e^{\beta}$ the spectrum is purely singular continuous for every phase. Strongly Liouville frequencies, those with $\beta(\alpha) > 0$, thus destroy localization even though the Lyapunov exponent stays positive.

What does a wave packet do in that window? This paper answers the question for every transport moment, and it determines the packing dimension of every spectral measure. All of these turn out to be one number.

1.1 Transport exponents

Let $\psi_t = e^{-itH}\delta_0$ and average in time with the exponential weight $(2/T)e^{-2t/T}$:

$$a(j,T) = \frac{2}{T}\int_0^\infty e^{-2t/T}\,|\langle \delta_j, \psi_t\rangle|^2\,dt, \qquad P(N,T) = \sum_{|j|>N} a(j,T), \qquad M_p(T) = \sum_{j\in\mathbb{Z}} |j|^p a(j,T). \tag{3}$$

$P(N,T)$ is the time-averaged probability of finding the particle beyond distance N, and $M_p(T)$ is the averaged p-th moment of the position. The upper transport exponents are

$$\beta^+(p) = \limsup_{T\to\infty} \frac{\log M_p(T)}{p\log T}, \qquad \alpha_u^+ = \sup_{p>0}\beta^+(p). \tag{4}$$

The quantity α_u^+ is the fastest upper exponent of Damanik and Tcheremchantsev [11]. It sees even polynomially small tails: $\alpha_u^+ \le a$ holds when $P(T^a,T)$ decays faster than every power of T, and conversely a tail of size T^{-C} at distance T^a already forces $\alpha_u^+ \ge a$. Always $0 \le \alpha_u^+ \le 1$, since hopping is bounded.

Two endpoints were known. Damanik and Tcheremchantsev proved $\alpha_u^+ = 0$ for $\lambda > 1$ when $\beta(\alpha) = 0$ [11], and Jitomirskaya and Zhang proved $\alpha_u^+ = 1$ for every phase when $\beta(\alpha) = \infty$ [18]. Jitomirskaya and Krüger proved exponential dynamical localization when $\beta(\alpha)$ is sufficiently small, including positive values [15], and Bradshaw, de Jong, Liu, Wang, Wang and Yang refined the growth rates for weakly Liouville frequencies with $\beta(\alpha) = 0$ [7]. In between, Jitomirskaya and Zhang obtained lower bounds of the form $1 - C\log\lambda/\beta$, for every moment [18]. Haeming refined them, with an explicit constant, a logarithmic loss and one sequence of times common to all phases [14]. Upper bounds on fractal dimensions of spectral measures near the transition were obtained by Jitomirskaya, Liu and Tcheremchantsev [17]. The exact value was open.

1.2 Main results

The answer depends on the phase only through a full-measure condition. Let

$$G_\alpha = \{\theta \in \mathbb{T} :\ |\sin(2\pi q_n\theta)| \ge q_n^{-2} \text{ for all sufficiently large } n\}. \tag{5}$$

This set has full Lebesgue measure for every irrational α (Lemma 2.1). Throughout, the number $\max\{0, 1 - \log\lambda/\beta(\alpha)\}$ is read as 0 when $\beta(\alpha) = 0$ and as 1 when $\beta(\alpha) = \infty$.

Theorem A (The transport law)**.** *Let $\lambda > 1$ and let α be irrational.*

1. *If $\beta(\alpha) < \infty$, then for every $\theta \in G_\alpha$ and every $p > 0$,*

$$\beta^+(p) = \alpha_u^+ = \max\Big\{0,\ 1 - \frac{\log\lambda}{\beta(\alpha)}\Big\}$$

   for $H_{\lambda,\alpha,\theta}$, where the right side is read as 0 when $\beta(\alpha) = 0$. Moreover, for every $a > \max\{0, 1 - \log\lambda/\beta(\alpha)\}$ and every $m > 0$, $P(T^a,T) = O(T^{-m})$ as $T\to\infty$.

2. *If $\beta(\alpha) = \infty$, then $\beta^+(p) = \alpha_u^+ = 1$ for every phase and every $p > 0$.*

The lower bound $\beta^+(p) \geq \max\{0, 1 - \log\lambda/\beta(\alpha)\}$ in part 1 holds for every phase and every $p > 0$.

Part 2 is due to Jitomirskaya and Zhang [18, Corollary 3], and the case $\beta(\alpha) = 0$ of part 1 to Damanik and Tcheremchantsev [11]; we include them so that the statement covers every frequency. The new content is part 1 for $0 < \beta(\alpha) < \infty$: the upper bound throughout, and the matching lower bound for every moment when $\beta(\alpha) > \log\lambda$. For the fastest exponent α_u^+ alone, the lower bound also holds for any Lipschitz sampling function whose Lyapunov exponent is continuous near a point of the spectrum where it is minimal, with $\log\lambda$ replaced by that minimum (Theorem 8.1).

Theorem A says that transport is governed by a competition between two exponential rates: the localization strength $\log\lambda$ and the Liouville strength $\beta(\alpha)$. The exponent vanishes for $\lambda \geq e^{\beta}$, including the transition point itself. When $\beta(\alpha) > 0$ it rises continuously to 1 as $\lambda \downarrow 1$; when $\beta(\alpha) = 0$ it vanishes for every $\lambda > 1$. For every moment and almost every phase, the arithmetic spectral transition of [2] is therefore also a sharp dynamical transition, and the theorem gives the exact exponents on the singular continuous side.

The formula has a simple heuristic origin. At a resonant scale $q = q_n$ the potential is almost q-periodic: a shift by q sites moves the phase by $\delta = q\alpha - p$, a drift of $|\delta| \approx e^{-\beta q}$ per period, so the periodic structure changes only over the slow spatial scale $|\delta|^{-1} \approx e^{\beta q}$. On that scale the packet moves like a particle in a nearly periodic band of width about λ^{-q}, so its speed is about $e^{-q\log\lambda}$. The slow drift turns this into a Bloch-type oscillation that lasts for a time about $T = e^{\beta q}$. In that time the packet covers a distance about $Te^{-q\log\lambda} = T^{1-\log\lambda/\beta}$. The theorem shows that no other mechanism moves any polynomial fraction of the packet faster, at any time. The next theorem shows that this mechanism moves a fixed fraction of the packet, not just a small tail. Write $\delta_n = q_n\alpha - p_n$ and $b_n = q_n^{-1}\log|\delta_n|^{-1}$, so that $\limsup_n b_n = \beta(\alpha)$.

Theorem B (A fixed fraction moves)**.** *Let $\lambda > 1$, let α be irrational and let θ be any phase. Let $h > \log\lambda$ and $c > h$. There is n_0 such that for every $n \geq n_0$ with $b_n \geq c$ and every T with $e^{hq_n} \leq T \leq |\delta_n|^{-1}$,*

$$P(Te^{-hq_n}, T) \geq \frac{2^{-32}}{1 + 4\pi\lambda}.$$

At the resonant time $T = |\delta_n|^{-1}$ the distance is $e^{(b_n - h)q_n} = T^{1-h/b_n}$. Along scales with $b_n \to \beta(\alpha)$ this gives the lower bound in Theorem A for every moment at once. The constant is crude, but it depends only on λ. For earlier times the theorem says that a fixed fraction is beyond Te^{-hq_n}. For $\theta \in G_\alpha$ and $\beta(\alpha) < \infty$, the window estimate of Section 5 makes the Abel-averaged mass beyond $2e^{\varepsilon q_n}T\lambda^{-q_n}$ exponentially small in q_n, for any fixed $\varepsilon > 0$. Together they show that at each strongly resonant scale and for $e^{hq_n} \leq T \leq |\delta_n|^{-1}$, the annulus $Te^{-hq_n} < |j| \leq 2e^{\varepsilon q_n}T\lambda^{-q_n}$ carries Abel-averaged mass at least $2^{-32}/(1 + 4\pi\lambda) - Ce^{-q_n}$ (Remark 9.8). So a fixed positive fraction of the packet lies at distance $T\lambda^{-q_n}$, up to factors $e^{(h-\log\lambda)q_n}$ and $2e^{\varepsilon q_n}$ whose exponents can be fixed as small as we like, and it moves out with speed λ^{-q_n} in this sense. The estimates say nothing about how the rest of the packet, up to mass $1 - 2^{-32}/(1 + 4\pi\lambda)$, is distributed within distance Te^{-hq_n}.

The same number governs the spectral measures. For a finite Borel measure μ on $\mathbb{R}$ and $E \in \mathbb{R}$, the upper local dimension is

$$d_\mu^+(E) = \limsup_{r\downarrow 0} \frac{\log\mu([E - r, E + r])}{\log r}.$$

The lower and upper packing dimensions $\dim_P^-(\mu)$ and $\dim_P^+(\mu)$ are its μ-essential infimum and supremum. Let μ_φ be the spectral measure of $\varphi \in \ell^2(\mathbb{Z})$ for $H_{\lambda,\alpha,\theta}$.

Theorem C (Packing dimension)**.** *Let $\lambda > 1$ and let α be irrational.*

1. *For every phase θ and every nonzero $\varphi \in \ell^2(\mathbb{Z})$, $d^+_{\mu_\varphi}(E) \geq \max\{0, 1 - \log\lambda/\beta(\alpha)\}$ for μ_φ-almost every E, with the right side read as 0 when $\beta(\alpha) = 0$ and as 1 when $\beta(\alpha) = \infty$. In particular $\dim^-_P(\mu_\varphi)$ is at least this number.*

2. *If $\beta(\alpha) < \infty$ and $\theta \in G_\alpha \cap (G_\alpha - \alpha)$, then for every nonzero $\varphi \in \ell^2(\mathbb{Z})$,*

$$d^+_{\mu_\varphi}(E) = \max\Big\{0,\ 1 - \frac{\log\lambda}{\beta(\alpha)}\Big\} \quad \text{for } \mu_\varphi\text{-a.e. } E,$$

so that $\dim^-_P(\mu_\varphi) = \dim^+_P(\mu_\varphi) = \max\{0, 1 - \log\lambda/\beta(\alpha)\}$, read as 0 when $\beta(\alpha) = 0$.

When $\beta(\alpha) = \infty$, part 1 gives $d^+_{\mu_\varphi} \geq 1$ almost everywhere at every phase. Every finite Borel measure on $\mathbb{R}$ has upper packing dimension at most 1, since this dimension is the infimum of the packing dimensions of its full-measure Borel sets [13, Appendix]. So all these packing dimensions equal 1; this endpoint is due to Jitomirskaya and Zhang [18, Corollary 3].

The set $G_\alpha \cap (G_\alpha - \alpha)$ has full measure (Lemma 2.1). So for this model the fastest-tail exponent, every moment exponent and the packing dimension coincide: if $\beta(\alpha) < \infty$ and $\theta \in G_\alpha \cap (G_\alpha - \alpha)$, then for all $p > 0$ and all nonzero φ,

$$\alpha^+_u = \beta^+(p) = \dim^-_P(\mu_\varphi) = \dim^+_P(\mu_\varphi) = \max\Big\{0,\ 1 - \frac{\log\lambda}{\beta(\alpha)}\Big\}.$$

For $1 < \lambda < e^\beta$, Jitomirskaya, Liu and Tcheremchantsev proved the bound $2(1 - \log\lambda/\beta)$ for every α-Diophantine phase, a full-measure set, and noted that the factor 2 may not be sharp [17, Theorem 1.1]. Theorem C shows that on $G_\alpha \cap (G_\alpha - \alpha)$ the sharp factor is 1, and that the bound is attained: the upper and lower packing dimensions both equal $1 - \log\lambda/\beta$. Their phase set and ours are both of full measure, but neither is shown to contain the other. For $\lambda \geq e^\beta$ their theorem already gives dimension zero. In the other direction, Jitomirskaya and Zhang proved lower bounds of the form $1 - C\log\lambda/\beta$ for packing dimensions [18]. Part 1 gives this bound with $C = 1$ at every phase, and part 2 shows that $C = 1$ cannot be improved.

The upper bound in part 2 follows from Theorem A and the inequality between moment growth and packing dimension of Guarneri and Schulz-Baldes [13], in the form of Barbaroux, Germinet and Tcheremchantsev [3]. Part 1 does not follow from transport bounds, since those inequalities go the other way. It comes from a pointwise bound on the resolvent at the resonant scales (Lemma 9.10), which the proof of Theorem B also provides.

The last result concerns computability. The exponent in Theorem A is an explicit function of $\beta(\alpha)$, and $\beta(\alpha)$ is an asymptotic quantity of the continued fraction: it can be noncomputable even when α is computable. A real x is *right-c.e. relative to* $\mathbf{0}'$ if the set of rationals $r > x$ is computably enumerable relative to the halting problem.

Theorem D (Computability)**.** 1. *Let $\lambda > 1$ be computable, let α be a computable irrational with $\beta(\alpha) < \infty$, and let $\theta \in G_\alpha$ be computable. Then $\alpha^+_u(H_{\lambda,\alpha,\theta})$ is computable from $\beta(\alpha)$. If moreover $\beta(\alpha) > \log\lambda$, then α^+_u and $\beta(\alpha)$ have the same Turing degree.*

2. *Take $\lambda = 2$ and $\theta = 1/4$. As α ranges over the computable irrationals whose convergent denominators are eventually odd, $\alpha^+_u(H_{2,\alpha,1/4})$ ranges over exactly the reals in $[0,1]$ that are right-c.e. relative to $\mathbf{0}'$.*

3. *There is an explicit computable irrational α such that $\alpha^+_u(H_{2,\alpha,1/4}) \in [11/23, 1/2]$ has Turing degree $\mathbf{0}''$.*

All Hamiltonian data in part 3 are computable: the coupling 2, the phase 1/4 and every digit of α. Yet the transport exponent of this explicit operator cannot be computed even with an oracle for the halting problem. Undecidability of physical quantities is known for families of Hamiltonians, most prominently the spectral gap of Cubitt, Perez-Garcia and Wolf [9]. Here a single explicit instance of a textbook one-dimensional model has an exponent of exactly known, high Turing degree. Part 2 identifies the exact set of values, the reals that are right-c.e. relative to $\mathbf{0}'$. All of them are computable from $\mathbf{0}''$, and part 3 shows that degree $\mathbf{0}''$ is attained.

1.3 How the proof works

The lower bound for α_u^+ (Section 8) constructs, at a resonant scale $q = q_n$ and for any h slightly above $\log\lambda$, a band of energies on which complex-energy transfer matrices stay bounded by e^{2hq} over a distance $N_q = e^{(b_n-h)q+O(\log q)}$, at times $T_q = e^{b_n q+O(\log q)}$. The resolvent then sends a polynomially small mass to distance $N_q/2$ at time T_q. It applies to any Lipschitz sampling function whose Lyapunov exponent is continuous near a spectral minimizer. A polynomially small mass does not determine any fixed moment; Section 9 replaces it by a fixed fraction.

The upper bound is the main work. Its key object is an exactly conserved quantity at each scale. Chambers' formula [8] says that the discriminant of the periodic approximant with frequency p/q depends on the phase only through a constant term $2\lambda^q\cos 2\pi q\theta$. Combined with the periodic magic formula of Damanik, Killip and Simon [10], this gives a degree-q polynomial P_q with

$$P_q(H) = V + K, \qquad V(j) = 2\lambda^q\cos\bigl(2\pi(q\theta + j\,\delta)\bigr), \qquad \delta = q\alpha - p,$$

where K has range q (Proposition 3.3). At every denominator with $|q\alpha - p| \le e^{-cq}$ for a fixed $c > 0$, K is exponentially small compared with λ^q (Lemma 3.4). The operator $P_q(H)$ commutes with H exactly, so no error accumulates in time. Its diagonal V is a cosine of amplitude $2\lambda^q$ that drifts slowly in space. Three estimates are built on it.

1. *Barriers* (Section 4). On the spectral subspace where $P_q(H)$ is near its initial value, the monotone part of the drifting cosine acts as a wall. A Sylvester-equation argument bounds the mass that crosses it, at all times. Dividing $P_q(H)$ by the size of K makes this work at every scale where $|q\alpha - p|$ is exponentially small, even where K itself is large.

2. *Exact slow dynamics* (Section 5). At scales with $|q\alpha - p| \le e^{-(\log\lambda+\eta)q}$, smooth filters on the inverse branches of P_q turn the energy into a smooth function of $P_q(H)/\lambda^q$. On each filtered piece the evolution is exactly generated by an operator whose position commutators are of size λ^{-q} up to subexponential factors. This gives the velocity λ^{-q} for all time.

3. *A per-denominator estimate* (Section 6). The complex-energy method of Damanik and Tcheremchantsev [11] confines the packet to distance $O(q)$ up to times $e^{\kappa q}$, at every denominator q.

The final step (Section 7) is a covering argument. For each large time T, look at the first denominator $q_j \ge K\log T$. If it is not too large, the per-denominator estimate confines the packet. If it is very large, the preceding denominator has a definite Liouville ratio and T lies in a fixed logarithmic window of that scale, where the barrier and slow-dynamics estimates apply. No assumption on how the ratios $\log q_{n+1}/q_n$ behave between these scales is needed.

The fixed fraction (Section 9) uses the conserved polynomial a second time. Near the origin, $P_q(H) - 2\lambda^q\cos 2\pi q\theta$ acts like $S^q + S^{-q}$, whose diagonal matrix elements are the moments of the arcsine law on $[-2, 2]$. So under the spectral measure of δ_0 the rational discriminant becomes arcsine

distributed (Lemma 9.1), and most of the spectral mass sits at energies where the first q-blocks on both sides of the origin are elliptic with a definite margin. At such an energy E, for a fixed $h > \log\lambda$ and $e^{hq} \le T \le |q\alpha - p|^{-1}$, the resolvent column $k \mapsto \langle \delta_k, (H - E - i/T)^{-1}\delta_0\rangle$ stays spread over a run of consecutive blocks that reaches beyond distance Te^{-hq}. Two elementary facts make this work for every polarization of the solution: the sum of the masses of two adjacent blocks is bounded below, uniformly in the polarization, by a constant times the squared norm of the solution in the rotation frame, weighted by the block sum S_0 of Lemma 9.5, although one block alone can carry little mass, and a flux identity bounds the mass beyond the run (Lemma 9.5). Parseval's formula turns a fixed fraction of the resolvent column into a fixed fraction of the packet. At the same energies, the runs bound $|\langle \delta_0, (H - E - i|q\alpha - p|)^{-1}\delta_0\rangle|$ by $e^{(\log\lambda + o(1))q}$, hence the spectral mass of intervals of length $|q\alpha - p|$, and this gives the lower bound on local dimensions.

1.4 Organization

Section 2 fixes notation and collects elementary facts. Section 3 constructs the conserved polynomial. Section 4 proves the barrier estimates and Section 5 the window theorem for slow dynamics. Section 6 records the per-denominator estimate. Section 7 proves the upper bound in Theorem A, and Section 8 a lower bound for α_u^+ that holds for Lipschitz potentials. Section 9 proves the fixed-fraction lower bound and Theorems A, B and C. Section 10 proves Theorem D. Section 11 discusses what remains open.

2 Preliminaries

2.1 Notation

Throughout, $\lambda > 1$ is fixed, $L = \log\lambda > 0$, $v(x) = 2\lambda\cos(2\pi x)$, and $H = H_{\lambda,\alpha,\theta}$ is the operator (1). We write S for the shift $(Su)(n) = u(n+1)$, X for the position operator $X\delta_j = j\delta_j$, and $\mathbf{1}_A$ for multiplication by the indicator of $A \subset \mathbb{Z}$. For a frequency $\omega \in \mathbb{R}$ and a phase φ we also use $H_{\lambda,\omega,\varphi}$ with ω rational. The spectrum Σ of $H_{\lambda,\alpha,\theta}$ does not depend on θ when α is irrational, since lattice translations conjugate $H_{\lambda,\alpha,\theta}$ to $H_{\lambda,\alpha,\theta+k\alpha}$, the orbit $\theta + k\alpha$ is dense, and the operator depends norm-continuously on the phase.

Transfer matrices. For $E \in \mathbb{C}$ and $x \in \mathbb{T}$ put $T(E,x) = \begin{pmatrix} E - v(x) & -1 \\ 1 & 0 \end{pmatrix}$ and, for $r \ge 0$,

$$A_r(E,x) = T(E, x + (r-1)\alpha)\cdots T(E,x), \qquad A_0 = I.$$

A solution of $Hu = Eu$ satisfies $(u(r), u(r-1))^{\mathsf{T}} = A_r(E,\theta)(u(0), u(-1))^{\mathsf{T}}$. For a frequency ω other than α we write $A_r^{(\omega)}$. The Lyapunov exponent is $L(E) = \lim_r r^{-1}\int_{\mathbb{T}} \log\|A_r(E,x)\|\,dx = \inf_r r^{-1}\int_{\mathbb{T}} \log\|A_r(E,x)\|\,dx$. For $\lambda > 1$ it equals $L = \log\lambda$ at every $E \in \Sigma$ [6, Corollary 2].

Continued fractions. Let p_n/q_n be the convergents of α and put

$$\delta_n = q_n\alpha - p_n, \qquad D_n = |\delta_n|^{-1}, \qquad \beta_n = \frac{\log q_{n+1}}{q_n}, \qquad b_n = \frac{\log D_n}{q_n}.$$

The classical inequalities $(q_{n+1} + q_n)^{-1} < |\delta_n| < q_{n+1}^{-1}$ give

$$q_{n+1} < D_n < 2q_{n+1}, \qquad 0 < b_n - \beta_n < \frac{\log 2}{q_n}. \tag{6}$$

In particular $\limsup b_n = \limsup \beta_n = \beta(\alpha)$. Denominators grow at least like Fibonacci numbers, and $|\alpha - p_n/q_n| < q_n^{-2}$. When the scale n is fixed we drop the index and write q, p, δ, b, and $A = \lambda^q$. We also write $s = |\sin(2\pi q\theta)|$.

All constants below may depend on λ, on α and on parameters that are explicitly fixed. They never depend on the phase θ, the scale n or the time T, unless this is indicated. Thresholds of the form "for all sufficiently large n" are likewise uniform in the phase in Sections 3–6, among phases satisfying the stated sine condition at that scale. In Section 7 the phase is fixed and thresholds may depend on it.

2.2 The phase set

Lemma 2.1. *For every irrational α the set G_α in (5) has full Lebesgue measure.*

Proof. The map $\theta \mapsto q_n\theta$ preserves Lebesgue measure on $\mathbb{T}$, and $\{x \in \mathbb{T} : |\sin 2\pi x| < u\}$ has measure at most u. Hence $\{\theta : |\sin 2\pi q_n\theta| < q_n^{-2}\}$ has measure at most q_n^{-2}. Since $\sum_n q_n^{-2} < \infty$, the Borel–Cantelli lemma shows that almost every θ lies in only finitely many of these sets. □

2.3 Moments

Lemma 2.2. *1. For every $p > 0$ there is C_p, depending only on p, such that $M_p(T) \le C_p(1+T)^p$ for all $T > 0$ and every bounded potential.*

2. For $0 < a < 1$ and $T \ge 1$, $M_p(T) \le T^{ap} + \big(M_{2p}(T)\, P(T^a, T)\big)^{1/2}$.

3. If $0 < a < 1$ and $P(T^a, T) = O(T^{-m})$ for every $m > 0$, then $\alpha_u^+ \le a$.

Proof. 1. Let $k \ge 1$ be an integer. We have $[X, H] = S^{-1} - S$ and $[X^k, H] = \sum_{i=0}^{k-1} X^i[X, H]X^{k-1-i}$. Moving the shifts to the right with $S^{\pm1}X = (X \pm 1)S^{\pm1}$ writes $[X^k, H]$ as $\sum_\pm c_\pm(X)S^{\pm1}$, where $c_\pm$ are polynomials of degree at most $k-1$ with coefficients bounded by a constant depending only on k. Since $X\delta_0 = 0$, Duhamel's formula $X^k\psi_t = -i\int_0^t e^{-i(t-s)H}[X^k, H]\psi_s\, ds$ and induction on k give $\|X^k\psi_t\| \le C_k(1 + |t|)^k$. For real $p > 0$ choose an integer k with $2k \ge p$. Jensen's inequality for the probability distribution $|\psi_t(j)|^2$ gives

$$\sum_j |j|^p|\psi_t(j)|^2 \le \Big(\sum_j |j|^{2k}|\psi_t(j)|^2\Big)^{p/(2k)} = \|X^k\psi_t\|^{p/k} \le C(1 + |t|)^p.$$

Averaging against $(2/T)e^{-2t/T}$ proves the claim.

2. Split $M_p(T)$ into $|j| \le T^a$ and $|j| > T^a$. The first part is at most T^{ap}, because $\sum_j a(j, T) = 1$. Apply the Cauchy–Schwarz inequality to the second part, for the probability measure $a(\cdot, T)$ on $\mathbb{Z}$.

3. By parts 1 and 2, $M_p(T) \le T^{ap} + C(1 + T)^pT^{-m/2}$. Taking $m \ge 2p$ gives $M_p(T) \le CT^{ap}$, so $\beta^+(p) \le a$ for every p. □

2.4 Uniform growth near the spectrum

Lemma 2.3. *For every $h > L$ there are a bounded open set $U_h \supset \Sigma$ and a constant C_h such that*

$$\|A_r(E, x)\| \le C_h e^{hr} \qquad \text{for all real } E \in U_h,\ x \in \mathbb{T},\ r \ge 0.$$

Proof. Fix $E_0 \in \Sigma$ and $h_1 = (L + h)/2$. Since $L(E_0) = L < h_1$, there is r_0 with

$$\frac{1}{r_0}\int_{\mathbb{T}} \log \|A_{r_0}(E_0, x)\|\, dx < h_1.$$

The function $F_E(x) = \log \|A_{r_0}(E,x)\|$ is jointly continuous, so the same strict inequality holds for E in a compact neighborhood N_0 of E_0. Rotation by $r_0\alpha$ is uniquely ergodic, so Birkhoff averages of the continuous functions F_E converge to $\int F_E$ uniformly in x; by equicontinuity of $\{F_E\}_{E\in N_0}$ they converge uniformly in $E \in N_0$ as well. Hence there is m_0 with $\sum_{k<m} F_E(x + kr_0\alpha) \le mr_0 h$ for all $m \ge m_0$, $x \in \mathbb{T}$ and $E \in N_0$. Writing a product of length $r = mr_0 + s$ with $0 \le s < r_0$ as m blocks of length r_0 and a remainder, submultiplicativity gives $\|A_r(E,x)\| \le Ce^{hr}$ on N_0. A finite cover of the compact set Σ by the interiors of such neighborhoods gives U_h and C_h. □

2.5 Rational spectra

For a rational p/q in lowest terms let $\Sigma_{p/q} = \bigcup_{\varphi\in\mathbb{T}} \sigma(H_{\lambda,p/q,\varphi})$.

Lemma 2.4. *Let H_0 be a bounded Schrödinger operator on $\ell^2(\mathbb{Z})$ with hopping one and real potential, and let $E \in \sigma(H_0)$. For every integer $M \ge 1$ there is a unit vector u supported in an interval of length $2M$ with $\|(H_0 - E)u\| \le 8/M$.*

Proof. By Weyl's criterion there is a finitely supported vector w with $\|(H_0 - E)w\| \le M^{-1}\|w\|$. For $k \in \mathbb{Z}$ let $\chi_k(n) = \sin(\pi(n - kM)/(2M))$ for $kM \le n \le (k+2)M$ and $\chi_k(n) = 0$ otherwise. Then $\sum_k \chi_k(n)^2 = 1$, because on $[kM,(k+1)M]$ the two windows χ_k and χ_{k-1} are a sine and a cosine of the same angle. Also $|\chi_k(n+1) - \chi_k(n)| \le \pi/(2M)$, and for each n at most three indices k have $\chi_k(n)$ or $\chi_k(n+1)$ nonzero. Since $([S,\chi_k]w)(n) = (\chi_k(n+1) - \chi_k(n))w(n+1)$, this gives $\sum_k \|[S^{\pm1},\chi_k]w\|^2 \le 3\pi^2(4M^2)^{-1}\|w\|^2$. Hence $\sum_k \|\chi_k w\|^2 = \|w\|^2$ and

$$\sum_k \|(H_0 - E)\chi_k w\|^2 \le 2\sum_k \|\chi_k(H_0 - E)w\|^2 + 2\sum_k \|[H_0,\chi_k]w\|^2 \le \frac{2 + 6\pi^2}{M^2}\,\|w\|^2.$$

Some k therefore has $\|(H_0 - E)\chi_k w\| \le (8/M)\|\chi_k w\|$. □

Lemma 2.5. *There is C such that for every rational p/q:*

1. *every point of $\Sigma_{p/q}$ is within $C|\alpha - p/q|^{1/2}$ of Σ, and every point of Σ is within $C|\alpha - p/q|^{1/2}$ of $\Sigma_{p/q}$;*
2. *for every φ, every point of $\Sigma_{p/q}$ is within $2\pi\lambda/q$ of $\sigma(H_{\lambda,p/q,\varphi})$.*

Proof. 1. Let $E \in \sigma(H_{\lambda,p/q,\varphi})$ and take u from Lemma 2.4, supported in $[c - M, c + M]$. Choose θ' with $\theta' + c\alpha = \varphi + cp/q$. For $|n - c| \le M$ the potentials of $H_{\lambda,\alpha,\theta'}$ and $H_{\lambda,p/q,\varphi}$ differ by at most $4\pi\lambda M|\alpha - p/q|$. Hence $\operatorname{dist}(E,\Sigma) \le 8/M + 4\pi\lambda M|\alpha - p/q|$, and $M = \lceil |\alpha - p/q|^{-1/2} \rceil$ gives the first claim. The second claim is proved in the same way with the roles exchanged.

2. Translation by k sites conjugates $H_{\lambda,p/q,\varphi}$ to $H_{\lambda,p/q,\varphi+kp/q}$, and $\{kp/q\}$ is the full grid $q^{-1}\mathbb{Z}/\mathbb{Z}$. Every phase φ' is within $1/(2q)$ of a grid translate of φ, and the potential is $4\pi\lambda$-Lipschitz in the phase. □

3 The conserved polynomial

3.1 Rational input

For a rational p/q in lowest terms, the operator $H_{\lambda,p/q,\varphi}$ is q-periodic. Its discriminant is $\Delta_\varphi(E) = \operatorname{tr} A_q^{(p/q)}(E,\varphi)$, and its spectrum is $\{E : |\Delta_\varphi(E)| \le 2\}$.

Lemma 3.1 (Chambers' formula [8])**.** *There is a real monic polynomial $P_{p/q}$ of degree q such that*

$$\Delta_\varphi(E) = P_{p/q}(E) - 2\lambda^q \cos(2\pi q\varphi) \qquad \text{for all } E \in \mathbb{C},\ \varphi \in \mathbb{T}.$$

Proof. Each factor $T(E, \varphi + kp/q)$ is a trigonometric polynomial of degree one in φ, so $\Delta_\varphi(E)$ is a trigonometric polynomial of degree at most q in φ. The shift $\varphi \mapsto \varphi + p/q$ cyclically permutes the factors and leaves the trace unchanged. Since $\gcd(p, q) = 1$, the trace is invariant under $\varphi \mapsto \varphi + 1/q$, so only the Fourier modes 0 and $\pm q$ survive. The coefficient of $e^{2\pi i q\varphi}$ comes only from the product of the diagonal entries and equals $(-\lambda)^q e^{2\pi i p\, q(q-1)/(2q)} = (-1)^{q+p(q-1)}\lambda^q$. This is $-\lambda^q$ in all cases: if q is odd then $(-1)^q = -1$ and $p(q-1)$ is even, and if q is even then p is odd and $(-1)^{p(q-1)} = -1$. The coefficient of $e^{-2\pi i q\varphi}$ is its complex conjugate. The constant mode is a monic polynomial of degree q in E with real coefficients. □

Lemma 3.2 (Periodic magic formula [10])**.** *Let H_{per} be a q-periodic Schrödinger operator on $\ell^2(\mathbb{Z})$ with hopping one and discriminant Δ. Then $\Delta(H_{\mathrm{per}}) = S^q + S^{-q}$.*

This is the elementary direction of [10, Theorem 3.1], proved there by Floquet decomposition (equations (3.5)–(3.6)). The hypotheses hold: the operator is bounded, two-sided, periodic and has positive off-diagonal entries. Combining the two lemmas,

$$P_{p/q}(H_{\lambda,p/q,\varphi}) = S^q + S^{-q} + 2\lambda^q \cos(2\pi q\varphi). \tag{7}$$

3.2 The irrational operator

Proposition 3.3. *Fix $h > L$. There is C_h such that for all sufficiently large n and every phase θ the following holds, with $q = q_n$, $p = p_n$, $\delta = \delta_n$ and $P_q = P_{p/q}$.*

1. *The operator $Q = P_q(H_{\lambda,\alpha,\theta})$ commutes with $H_{\lambda,\alpha,\theta}$ and decomposes as*

$$Q = V + K, \qquad V(j) = 2\lambda^q \cos\bigl(2\pi(q\theta + j\delta)\bigr), \qquad K = S^q + S^{-q} + R,$$

where V is diagonal, R is self-adjoint, $\langle \delta_i, R\,\delta_j\rangle = 0$ for $|i - j| > q$, and

$$\|R\| \le C_h\, q^{5/2} e^{hq} |\delta|.$$

2. *For every fixed integer $r \ge 0$ there is $C_{r,h}$ with $\sup_{E\in U_h} |P_q^{(r)}(E)| \le C_{r,h}\, q^r e^{hq}$, where U_h is the neighborhood of Lemma 2.3.*

Proof. Step 1: rational products. Let $d = 4\pi\lambda|\delta|$. For $0 \le k < q$, the potentials $v(\varphi + kp/q)$ and $v(\varphi + k\alpha)$ differ by at most $4\pi\lambda k|\alpha - p/q| \le d$. Write each rational transfer matrix as the irrational one plus an error of norm at most d, and expand the product of $r \le q$ factors. A term with k errors contains at most $k + 1$ maximal irrational segments, each bounded by Lemma 2.3. Hence for real $E \in U_h$ and every phase,

$$\|A_r^{(p/q)}(E, \varphi)\| \le \sum_{k=0}^{r} \binom{r}{k} C_h^{k+1} e^{h(r-k)} d^k = C_h e^{hr}\bigl(1 + C_h d e^{-h}\bigr)^r \le C_h' e^{hr}, \tag{8}$$

because $rd \le 4\pi\lambda q|\delta| \le 4\pi\lambda$. The same bound holds for every sub-block. Differentiating $A_q^{(p/q)}(E, \varphi)$ r times in E inserts the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ at r positions. There are at most q^r ordered choices, and

each resulting product is bounded by $C^{r+1}e^{hq}$ by (8). Since $P_q(E) = \Delta_\varphi(E) + 2\lambda^q \cos(2\pi q\varphi)$ and $\lambda^q \le e^{hq}$, this proves part 2.

Step 2: a smooth cutoff. Let U' be an open set with $\Sigma \subset U' \subset \overline{U'} \subset U_h$, and fix a smooth compactly supported χ with $\chi = 1$ on U' and $\operatorname{supp}\chi \subset U_h$. Put $f_q = \chi P_q$. By part 2, $\|f_q\|_{H^2(\mathbb{R})} \le Cq^2e^{hq}$. For bounded self-adjoint operators B, C, Fourier inversion and $\|e^{itB} - e^{itC}\| \le |t|\,\|B - C\|$ give

$$\|f_q(B) - f_q(C)\| \le \frac{\|B-C\|}{2\pi}\int_{\mathbb{R}} |t|\,|\hat f_q(t)|\,dt \le c\,\|B-C\|\,\|f_q\|_{H^2(\mathbb{R})},$$

by the Cauchy–Schwarz inequality and $\int t^2(1+t^2)^{-2}\,dt < \infty$.

Step 3: rows. Fix $j \in \mathbb{Z}$. Let $\varphi_j = \theta + j(\alpha - p/q)$ and $B_j = H_{\lambda,p/q,\varphi_j}$, whose potential at site k is $v(\theta + j\alpha + (k-j)p/q)$. Let C_j be B_j with its potential on $[j-q, j+q]$ replaced by that of $H_{\lambda,\alpha,\theta}$. Those potentials differ by at most $4\pi\lambda|k-j|\,|\alpha - p/q| \le d$, so $\|B_j - C_j\| \le d$. By Lemma 2.5, $\sigma(B_j) \subset \Sigma_{p/q}$ lies within $C|\alpha - p/q|^{1/2}$ of Σ, and $\sigma(C_j)$ lies within d of $\sigma(B_j)$. For all large n both spectra lie in U', uniformly in j and θ, so $f_q(B_j) = P_q(B_j)$ and $f_q(C_j) = P_q(C_j)$.

A path of length at most q starting at j stays in $[j-q, j+q]$, so the j-th row of $P_q(C_j)$ equals the j-th row of $P_q(H_{\lambda,\alpha,\theta})$. By (7) the j-th row of $P_q(B_j)$ equals the j-th row of $S^q + S^{-q} + 2\lambda^q\cos(2\pi q\varphi_j)$, and $q\varphi_j = q\theta + j(q\alpha - p) = q\theta + j\delta$. Hence the j-th row of $R = Q - V - S^q - S^{-q}$ has ℓ^2 norm at most $\|f_q(C_j) - f_q(B_j)\| \le Cq^2e^{hq}|\delta|$. It has at most $2q+1$ nonzero entries, so its ℓ^1 norm is at most $\sqrt{2q+1}$ times that. Since Q, V and $S^q + S^{-q}$ are self-adjoint, so is R, and the Schur test gives $\|R\| \le C_h q^{5/2}e^{hq}|\delta|$. Finally Q is a polynomial in H, so it commutes with H. □

Only rows are compared in Step 3. The full irrational and rational operators are not close in norm, and R need not have a sign.

The proposition gives two regimes. When $b_n \ge L + \eta$ for a fixed $\eta > 0$, choosing $h < L + \eta$ makes $\|R\| \to 0$, so $\|K\| \le 3$ eventually. At all scales, the useful quantity is the size of K relative to λ^q:

$$\|K\| \le J_n := 3 + C_h q^{5/2}e^{hq}|\delta|, \qquad \sigma_n := \frac{J_n}{\lambda^q}. \tag{9}$$

Lemma 3.4. *Fix $c > 0$ and $0 < \xi < \min(L, c)/2$, and use $h = L + \xi$ in (9). Then $|\delta_n| \le \sigma_n \le e^{-(\min(L,b_n)-2\xi)q_n}$ for all sufficiently large n with $b_n \ge c$.*

Proof. Enlarging C_h, we may assume $C_h \ge 1$. Then $\sigma_n \ge C_h q^{5/2}e^{\xi q}|\delta| \ge |\delta|$. For the upper bound, $3\lambda^{-q} = 3e^{-Lq}$ and $C_h q^{5/2}e^{hq}|\delta|\lambda^{-q} = C_h q^{5/2}e^{\xi q}e^{-b_n q}$. Each is at most $\frac{1}{2}e^{-(\min(L,b_n)-2\xi)q}$ once q is large, uniformly in $b_n \ge c$. □

Thus K may be exponentially large when $b_n < L$, but it is exponentially small compared with λ^q at every scale with $b_n \ge c$.

4 Spectral barriers

4.1 An abstract barrier lemma

In this subsection $Q = V + K$ is a bounded self-adjoint operator on $\ell^2(\mathbb{Z})$, V is a real diagonal operator, K has range at most q (that is, $\langle\delta_i, K\delta_j\rangle = 0$ for $|i-j| > q$), and $\|K\| \le J$. Fix $d_0 \in \mathbb{R}$ and $W > 0$, and let $\Pi = \mathbf{1}_{[d_0-W, d_0+W]}(Q)$. For an interval $I \subset \mathbb{Z}$ write $\partial_q I = \{i \in I : \operatorname{dist}(i, \mathbb{Z}\setminus I) \le q\}$.

Lemma 4.1 (Outer barrier)**.** *Let $I \subset \mathbb{Z}$ be an interval on which either $V - d_0 \ge V_0$ or $V - d_0 \le -V_0$, where $V_0 \ge W + 2J$. If $B \subset I$ and $\operatorname{dist}(B, \mathbb{Z}\setminus I) > (\ell+1)q$ for an integer $\ell \ge 0$, then*

$$\|\mathbf{1}_B\Pi\| \le 2\Big(\frac{J}{V_0 - W}\Big)^{\ell+2}.$$

Proof. Assume $V - d_0 \geq V_0$ on I; the other case follows by replacing Q and d_0 with $-Q$ and $-d_0$. Let $Q_I = \mathbf{1}_I Q \mathbf{1}_I$ acting on $\ell^2(I)$, let A be the restriction of Q to $\operatorname{Ran}\Pi$, and let $Y = -\mathbf{1}_I Q \mathbf{1}_{\mathbb{Z}\setminus I}\Pi$. Since Π commutes with Q, the operator $X = \mathbf{1}_I\Pi$ solves the Sylvester equation

$$(Q_I - d_0)X - X(A - d_0) = Y.$$

Here $Q_I - d_0 \geq V_0 - J$ and $\|A - d_0\| \leq W < V_0 - J$, so

$$X = \int_0^\infty e^{-s(Q_I - d_0)}\, Y\, e^{s(A-d_0)}\, ds,$$

as one checks by differentiating the integrand. Since V is diagonal, $Y = -\mathbf{1}_I K \mathbf{1}_{\mathbb{Z}\setminus I}\Pi$, so $\|Y\| \leq J$ and the range of Y lies in $\ell^2(\partial_q I)$.

Expand $e^{-s(Q_I - d_0)}$ in a Dyson series around the diagonal part $V_I - d_0 \geq V_0$. The term with k factors of $K_I = \mathbf{1}_I K \mathbf{1}_I$ has norm at most $e^{-sV_0}(sJ)^k/k!$ and moves supports by at most kq. Since $\operatorname{dist}(B, \partial_q I) > \ell q$, only terms with $k \geq \ell + 1$ connect $\ell^2(\partial_q I)$ to $\ell^2(B)$. Therefore

$$\|\mathbf{1}_B X\| \leq J \int_0^\infty e^{-s(V_0 - W)} \sum_{k \geq \ell+1} \frac{(sJ)^k}{k!}\, ds = \sum_{k \geq \ell+1} \Big(\frac{J}{V_0 - W}\Big)^{k+1} \leq 2\Big(\frac{J}{V_0 - W}\Big)^{\ell+2},$$

using $J/(V_0 - W) \leq \frac{1}{2}$. □

Lemma 4.2 (Inner core)**.** *Let $I_0 \subset \mathbb{Z}$ be an interval with $|V - d_0| \leq W/4$ on I_0, and assume $W \geq 4J$. If $0 \in I_0$ and $\operatorname{dist}(0, \mathbb{Z} \setminus I_0) > (\ell + 1)q$, then*

$$\|(1 - \Pi)\delta_0\| \leq 3\Big(\frac{4J}{3W}\Big)^{\ell+2}.$$

Proof. Let $\Pi_+ = \mathbf{1}_{(d_0 + W, \infty)}(Q)$ and let A_+ be the restriction of Q to its range, so $A_+ - d_0 \geq W$. With $Q_{I_0} = \mathbf{1}_{I_0} Q \mathbf{1}_{I_0}$ we have $Q_{I_0} - d_0 \leq W/4 + J$, and $X_+ = \mathbf{1}_{I_0}\Pi_+$ solves $(Q_{I_0} - d_0)X_+ - X_+(A_+ - d_0) = Y_+$ with $Y_+ = -\mathbf{1}_{I_0} K \mathbf{1}_{\mathbb{Z}\setminus I_0}\Pi_+$. Hence

$$X_+ = -\int_0^\infty e^{s(Q_{I_0} - d_0)}\, Y_+\, e^{-s(A_+ - d_0)}\, ds.$$

Expanding $e^{s(Q_{I_0} - d_0)}$ around its diagonal part, which has norm at most $W/4$, only terms with at least $\ell + 1$ factors of K connect $\partial_q I_0$ to the site 0. As in Lemma 4.1,

$$\|\Pi_+\delta_0\| = \|\delta_0^* X_+\| \leq J \int_0^\infty e^{-3sW/4} \sum_{k \geq \ell+1} \frac{(sJ)^k}{k!}\, ds \leq 2\Big(\frac{4J}{3W}\Big)^{\ell+2}.$$

The same bound holds for $\Pi_- = \mathbf{1}_{(-\infty, d_0 - W)}(Q)$, applying the argument to $-Q$. Since $1 - \Pi = \Pi_+ + \Pi_-$ with orthogonal ranges, the claim follows. □

Lemma 4.3 (Cutoff)**.** *Let Π be a spectral projection of an operator that commutes with H, let $N \geq 1$ be an integer, and put $B_N = \{-N - 1, -N, N, N + 1\}$. Then for every $T > 0$,*

$$P(N, T) \leq 8\|(1 - \Pi)\delta_0\|^2 + T^2\|\mathbf{1}_{B_N}\Pi\|^2.$$

Proof. Let $\chi = \mathbf{1}_{[-N,N]}$. The commutator $[H,\chi] = [S + S^{-1}, \chi]$ is supported on the two bonds $\{N, N+1\}$ and $\{-N-1, -N\}$, has norm one, and satisfies $[H,\chi] = [H,\chi]\mathbf{1}_{B_N}$. Since Π commutes with H, Duhamel's formula gives

$$(1-\chi)e^{-itH}\Pi\delta_0 = e^{-itH}(1-\chi)\Pi\delta_0 - i\int_0^t e^{-i(t-s)H}[H,\chi]\,\Pi\, e^{-isH}\delta_0\, ds.$$

As $(1-\chi)\delta_0 = 0$, we have $\|(1-\chi)\Pi\delta_0\| \le \|(1-\Pi)\delta_0\|$. Hence

$$\|(1-\chi)\psi_t\| \le 2\|(1-\Pi)\delta_0\| + |t|\,\|\mathbf{1}_{B_N}\Pi\|.$$

Square, use $(x+y)^2 \le 2x^2+2y^2$, and average with $(2/T)e^{-2t/T}$, for which the mean of t^2 is $T^2/2$. □

The time factor T cannot be removed. Mass that crosses the barrier does so at a small but nonzero rate, and the estimate only shows that the rate is tiny.

4.2 The local barrier at a scale

We now apply these lemmas to $Q = P_q(H)$ from Proposition 3.3, divided by J_n.

Proposition 4.4. *Fix $c > 0$ and $0 < \varepsilon < \min(L,c)/4$. There are C and n_0 such that the following holds for every $n \ge n_0$ with $b_n \ge c$ and every phase θ with $|\sin(2\pi q_n\theta)| \ge q_n^{-2}$. There is an integer N_n with*

$$e^{\varepsilon q_n/2} \le N_n \le 2e^{((b_n-L)_+ + \varepsilon)q_n} \qquad \text{and} \qquad P(N_n,T) \le C(1+T^2)\exp\Big(-\frac{\varepsilon N_n}{900}\Big) \quad \text{for all } T>0.$$

Proof. Write $q = q_n$, $\delta = \delta_n$, $b = b_n$, $A = \lambda^q$, $s = |\sin 2\pi q\theta| \ge q^{-2}$ and $\mu = \min(L,b) \ge \mu_0 = \min(L,c)$. Take $\xi = \varepsilon/8$ in Lemma 3.4, so that $|\delta| \le \sigma \le e^{-(\mu-\varepsilon/4)q}$ with $\sigma = J_n/A$. Put

$$N = \Big\lceil e^{\varepsilon q/2}\,\frac{\sigma}{|\delta|}\Big\rceil.$$

Then $N \ge e^{\varepsilon q/2}$, and $N \le 2e^{(b-\mu+3\varepsilon/4)q} \le 2e^{((b-L)_+ + \varepsilon)q}$, because $b - \mu = (b-L)_+$.

The drifting diagonal. We have $2\pi|\delta|N \le 2\pi(e^{\varepsilon q/2}\sigma + |\delta|) \le e^{-\varepsilon q}$ for large q, since $\mu_0 - \varepsilon/4 - \varepsilon/2 > \varepsilon$. For large q this is at most $s/50$. Write $x = 2\pi q\theta$ and $y_j = 2\pi j\delta$. For $|j| \le N$ the mean value theorem gives

$$V(j) - V(0) = 2A\big(\cos(x+y_j) - \cos x\big) = -2A\, y_j \sin(x + \vartheta_j y_j), \qquad \vartheta_j \in (0,1),$$

and $|\sin(x+\vartheta_j y_j) - \sin x| \le |y_j| \le s/50$. So $\sin(x+\vartheta_j y_j)$ has the sign of $\sin x$ and modulus between $0.98s$ and $1.02s$. Consequently $V(j) - V(0)$ has a fixed sign on $I_+ = [N/2, N]$, the opposite fixed sign on $I_- = [-N, -N/2]$, and on both intervals

$$|V(j) - V(0)| \ge V_0 := A|\delta|Ns.$$

On $I_0 = [-r, r]$ with $r = \lfloor N/500 \rfloor$ we have $|V(j) - V(0)| \le 4\pi \cdot 1.02\, A|\delta|rs \le V_0/32$.

Normalization. Apply the lemmas to $Q' = Q/J_n = V' + K'$, with $V' = V/J_n$ and $\|K'\| \le 1$. The operator Q' commutes with H, and its spectral projections are those of Q. Put $d_0 = V'(0)$, $V_0' = V_0/J_n$ and $W' = V_0'/8$. By the choice of N,

$$V_0' = \frac{A|\delta|Ns}{J_n} \ge \frac{A|\delta|s}{J_n}\cdot\frac{e^{\varepsilon q/2}J_n}{A|\delta|} = e^{\varepsilon q/2}s \ge e^{\varepsilon q/2}q^{-2}.$$

In particular $\log(7V_0'/8) \ge \varepsilon q/3$ for large q, and the hypotheses $V_0' \ge W' + 2$ and $W' \ge 4$ hold.

Outer barriers. Let $R = \lfloor 3N/4 \rfloor$. The sites $R, R+1$ lie in I_+ at distance at least $N/4 - 2$ from its complement, and similarly for $-R-1, -R$ in I_-. Lemma 4.1 with $\ell = \lfloor (N/4-2)/q \rfloor - 2 \ge N/(5q)$ gives

$$\|\mathbf{1}_{B_R}\Pi\| \le 4\Big(\frac{8}{7V_0'}\Big)^{\ell} \le 4\exp\Big(-\frac{N}{5q}\cdot\frac{\varepsilon q}{3}\Big) = 4e^{-\varepsilon N/15}.$$

Initial leakage. On I_0 we have $|V' - d_0| \le V_0'/32 = W'/4$. Lemma 4.2 with $\ell_0 = \lfloor r/q \rfloor - 2 \ge N/(600q)$ gives

$$\|(1-\Pi)\delta_0\| \le 3\Big(\frac{4}{3W'}\Big)^{\ell_0} \le 3\Big(\frac{32}{3V_0'}\Big)^{\ell_0} \le 3e^{-\varepsilon N/1800}.$$

Conclusion. Lemma 4.3 gives $P(R,T) \le 72e^{-\varepsilon N/900} + 16T^2 e^{-2\varepsilon N/15}$. Since $R \le N$ and $P(\cdot,T)$ is nonincreasing, the same bound holds for $P(N,T)$. All thresholds depend only on $\lambda, \alpha, c, \varepsilon$. Take $N_n = N$. □

Two features matter later. The physical evolution is never rescaled; only Q is divided by J_n, which changes neither its spectral projections nor its commutation with H. And the proposition needs only $b_n \ge c$: at scales with $b_n < L$ the operator K is large, but it is small compared with λ^q.

5 Exact slow dynamics at strongly resonant scales

This section treats scales where $|\delta_n|$ is much smaller than λ^{-q_n}, that is, $b_n \ge L + \eta$ for a fixed $\eta > 0$. On such scales the conserved polynomial gives an exact description of the dynamics on most of the spectral measure of δ_0, with velocity λ^{-q}.

Throughout the section $q = q_n$, $A = \lambda^q$, $s = |\sin 2\pi q\theta| \ge q^{-2}$, and $F = P_q/A$. Put

$$y_0 = 2\cos(2\pi q\theta), \qquad d = 2 - |y_0|, \qquad \rho = d/16, \qquad J_{\mathrm{big}} = [y_0 - 4\rho,\, y_0 + 4\rho].$$

Since $1 - |\cos x| \ge \frac{1}{2}\sin^2 x$ and $1 - |\cos x| \le \sin^2 x$, we have $q^{-4} \le s^2 \le d \le 2s^2$. We fix $h > L$ and write $E_h = e^{(h-L)q}$. By Proposition 3.3, $|F^{(k)}| \le C_{k,h} q^k E_h$ on U_h for each fixed k. By Lemma 2.5, $\Sigma_{p/q} \subset U_h$ for all large n, and we assume this.

5.1 Inverse branches and filters

Lemma 5.1. *For all large n and every phase with $s \ge q^{-2}$:*

1. *For each $y \in (-2,2)$ the equation $F(E) = y$ has q simple real roots, all in $\Sigma_{p/q}$. They define q smooth monotone branches $e_1(y) < \cdots < e_q(y)$ on J_{big}.*
2. *For $b \ne b'$ the arcs $e_b(J_{\mathrm{big}})$ and $e_{b'}(J_{\mathrm{big}})$ are at distance at least $12\rho/(C_{1,h} q E_h)$.*
3. *$|e_b'(y)| \le 2\lambda/(qs)$ on J_{big}, and for each $k \ge 1$ there are $C_{k,h}$ and c_k, depending only on k, with $|e_b^{(k)}(y)| \le C_{k,h}\, q^{c_k} E_h^{k-1}$ on J_{big}.*

Proof. 1. Let $y = 2\cos(2\pi q\varphi)$. By Lemma 3.1, $F(E) = y$ exactly when $\Delta_\varphi(E) = 0 = 2\cos(\pi/2)$, that is, when E is an eigenvalue of the Floquet fiber of $H_{\lambda,p/q,\varphi}$ with Bloch phase $\pi/2$. This fiber is a $q \times q$ Hermitian matrix, so there are q real roots counted with multiplicity, all in $\sigma(H_{\lambda,p/q,\varphi}) \subset \Sigma_{p/q}$. A double root would give two independent Bloch solutions with multiplier i, forcing the real unimodular monodromy matrix to equal iI, which is impossible. Since the roots are simple, the implicit function theorem gives the smooth branches.

2. Simplicity for every $y \in (-2,2)$ means that every critical value of P_q has modulus at least $2A$. Between the arcs $e_b(J_{\text{big}})$ and $e_{b+1}(J_{\text{big}})$ there is exactly one critical point E_c of P_q, because $P_q - Ay$ has q simple real roots for each $y \in J_{\text{big}}$ and P_q' has degree $q-1$. Let E_1 be the endpoint of $e_b(J_{\text{big}})$ facing E_c, with $F(E_1) = y_1 \in J_{\text{big}}$. On $[E_1, E_c]$ the function P_q is monotone, and it moves from Ay_1 to a value of modulus at least $2A$. Let E' be the first point where $|P_q| = 2A$. On $[E_1, E']$ we have $|P_q| \le 2A$, so this interval lies in $\Sigma_{p/q} \subset U_h$, where $|P_q'| \le C_{1,h} q e^{hq}$. Since $2 - |y_1| \ge d - 4\rho = 12\rho$, the change of P_q over $[E_1, E']$ is at least $12A\rho$, so $|E' - E_1| \ge 12A\rho/(C_{1,h} q e^{hq})$.

3. Let $E = e_b(y)$ and $y = 2\cos(2\pi q\varphi)$. The eigenvalue $E(\varphi)$ of the Bloch-phase-$\pi/2$ fiber is simple, and differentiating $\Delta_\varphi(E(\varphi)) = 0$ gives $P_q'(E)E'(\varphi) = -4\pi q A \sin(2\pi q\varphi)$. By the Hellmann–Feynman formula, $|E'(\varphi)|$ is at most the norm of the phase derivative of the fiber, which is at most $4\pi\lambda$. Hence

$$|P_q'(E)| \ge \frac{qA|\sin(2\pi q\varphi)|}{\lambda}. \tag{10}$$

For $y \in J_{\text{big}}$, $\sin^2(2\pi q\varphi) = 1 - y^2/4 \ge (2-|y|)/2 \ge 6\rho \ge 3s^2/8$, so $|\sin(2\pi q\varphi)| \ge s/2$ and $|e_b'(y)| = 1/|F'(E)| \le 2\lambda/(qs) \le 2\lambda q$. Differentiating $F(e_b(y)) = y$ repeatedly expresses $F'(e_b)e_b^{(k)}$ as a sum of products of $F^{(j)}(e_b)$, $2 \le j \le k$, with lower derivatives of e_b whose orders add up to k. Each such product contains exactly one factor $F^{(j)}$, bounded by $Cq^j E_h$, and induction on k gives the stated bound. □

Fix a smooth $f_0 : \mathbb{R} \to [0,1]$ with $f_0 = 1$ on $[-1,1]$ and $f_0 = 0$ outside $[-2,2]$, and put $f(y) = f_0((y-y_0)/\rho)$. For each branch b choose a smooth $\chi_b : \mathbb{R} \to [0,1]$ equal to 1 on $e_b([y_0 - 2\rho, y_0 + 2\rho])$ and supported in $e_b((y_0 - 3\rho, y_0 + 3\rho))$. By Lemma 5.1, the transition regions have length at least $\rho/\max|F'| \ge \rho/(C_{1,h} q E_h)$, and the supports of different χ_b are disjoint. Define

$$g_b(E) = \chi_b(E)\, f(F(E)), \qquad G_b(y) = \kappa(y)\, e_b(y),$$

where κ is a smooth cutoff equal to 1 on $[y_0 - 2\rho, y_0 + 2\rho]$ and supported in $[y_0 - 3\rho, y_0 + 3\rho]$, with $|\kappa^{(k)}| \le C_k \rho^{-k}$.

Lemma 5.2. *1. $\sum_b g_b(E) = f(F(E))$ for every real E.*

2. $G_b(F(E)) = E$ for every $E \in \operatorname{supp} g_b$.

3. For every $r \ge 0$ there are $C_{r,h}$ and c_r with $\|g_b\|_{C^r} + \|G_b\|_{C^r} \le C_{r,h} q^{c_r} E_h^r$. All supports lie in a fixed bounded interval.

4. For every $r \ge 1$, $\|X^r g_b(H)\delta_0\| \le C_{r,h}\, q^{c_{r+2}} E_h^{r+2}$.

Proof. 1. If $f(F(E)) \ne 0$ then $F(E) \in (y_0 - 2\rho, y_0 + 2\rho)$, so by Lemma 5.1 $E = e_b(F(E))$ for exactly one b. On that arc $\chi_b = 1$ and $\chi_{b'} = 0$ for $b' \ne b$.

2. On $\operatorname{supp} g_b$, E lies on the b-th arc and $F(E) \in \operatorname{supp} f$, where $\kappa = 1$; hence $G_b(F(E)) = e_b(F(E)) = E$.

3. This follows from $|f^{(k)}| \le C_k\rho^{-k} \le C_k 16^k q^{4k}$, the bounds on $F^{(k)}$ on U_h, the length of the transition regions of χ_b, and Lemma 5.1(3).

4. Since $X\delta_0 = 0$, $X^r g_b(H)\delta_0 = \operatorname{ad}_X^r(g_b(H))\delta_0$. Write $g(H) = (2\pi)^{-1}\int \hat g(t) e^{itH}\,dt$ and $H_u = e^{iuX} H e^{-iuX}$. Then $\operatorname{ad}_X^r(e^{itH}) = (-i\partial_u)^r e^{itH_u}|_{u=0}$, and all u-derivatives of H_u have norm at most 2. Repeated Duhamel expansion gives $\|\operatorname{ad}_X^r(e^{itH})\| \le C_r(1+|t|)^r$. Hence $\|\operatorname{ad}_X^r(g(H))\| \le C_r \int (1+|t|)^r |\hat g(t)|\,dt \le C_r \|g\|_{C^{r+2}}$ for g supported in a fixed bounded interval. Apply part 3. □

5.2 The exact slow propagator

Proposition 5.3. *Assume $\|K\| \le 3$ and $9q^{r+1} \le A$. Put $B = Q/A = F(H)$ and $H_b = G_b(B)$. Then:*

1. *$e^{-itH} g_b(H) = e^{-itH_b} g_b(H)$ for every $t \in \mathbb{R}$.*
2. *$\|\operatorname{ad}_X^k B\| \le 3(2q+1)q^k/A$ for every $k \ge 1$.*
3. *$\|\operatorname{ad}_X^k H_b\| \le C_k\, q^{k+1} A^{-1} \int (1+|t|)^k |\hat{G}_b(t)|\, dt$ for $1 \le k \le r$.*
4. *If $\gamma \ge \max_{1\le k\le r} \|\operatorname{ad}_X^k H_b\|$ and $M_0 \ge \max_{0\le k\le r} \|X^k g_b(H)\delta_0\|$, then*

$$\|X^r e^{-itH_b} g_b(H)\delta_0\| \le C_r M_0 (1+\gamma|t|)^r \qquad \text{for all } t.$$

Proof. 1. By Lemma 5.2(2), $\phi(E) g_b(E) = \phi(G_b(F(E)))\, g_b(E)$ for every function ϕ and every real E. Taking $\phi(E) = e^{-itE}$ and applying the functional calculus of H gives the identity, since $G_b(F(H)) = H_b$. No error term appears: $B = P_q(H)/A$ includes the residual R exactly.

2. Write $K = \sum_{m=-q}^{q} K_m$, where K_m keeps the entries $\langle \delta_i, K\delta_j\rangle$ with $i - j = m$. Each

$$K_m = \frac{1}{2\pi} \int_0^{2\pi} e^{-imu}\, e^{iuX} K e^{-iuX}\, du$$

has $\|K_m\| \le \|K\|$, and $\operatorname{ad}_X K_m = m K_m$. The diagonal V commutes with X. Hence $\|\operatorname{ad}_X^k B\| = \|\operatorname{ad}_X^k K\|/A \le 3(2q+1)q^k/A$.

3. Write $G_b(B) = (2\pi)^{-1} \int \hat{G}_b(t) e^{itB}\, dt$ and $B_u = e^{iuX} B e^{-iuX}$, so that $\|\partial_u^j B_u\| = \|\operatorname{ad}_X^j B\| \le 9q^{j+1}/A \le 1$. The derivative $\partial_u^k e^{itB_u}$ is a sum, over ordered partitions of $\{1,\dots,k\}$ into blocks, of time-ordered integrals containing one factor $\partial_u^{|\beta|} B_u$ per block β. A partition into m blocks contributes at most $|t|^m \prod_\beta \|\partial_u^{|\beta|} B_u\| \le |t|^m\, 9q^{k+1}/A$, since every factor is at most 1 and one factor is at most $9q^{k+1}/A$. Summing gives $\|\operatorname{ad}_X^k e^{itB}\| \le C_k (1+|t|)^k q^{k+1}/A$, and integrating against $\hat{G}_b$ proves part 3.

4. Let $\phi_t = e^{-itH_b} g_b(H)\delta_0$. The identity $X^k Y = \sum_{j=0}^{k} \binom{k}{j} (\operatorname{ad}_X^j Y) X^{k-j}$ gives $[X^k, H_b] = \sum_{j=1}^{k} \binom{k}{j} (\operatorname{ad}_X^j H_b) X^{k-j}$. Duhamel's formula then yields

$$\|X^k \phi_t\| \le \|X^k \phi_0\| + \gamma \sum_{j=1}^{k} \binom{k}{j} \int_0^{|t|} \|X^{k-j} \phi_\tau\|\, d\tau,$$

and induction on $k \le r$ gives the claim. Truncating X to bounded regions and passing to strong limits justifies these manipulations. □

5.3 The window theorem

Theorem 5.4. *Fix $\eta > 0$, $\varepsilon > 0$, $\Lambda \ge 1$ and $M > 0$. There are C and n_0 such that for every $n \ge n_0$ with $b_n \ge L + \eta$, every phase with $|\sin 2\pi q_n \theta| \ge q_n^{-2}$, and every $1 \le T \le e^{\Lambda q_n}$,*

$$P(R_n(T), T) \le C e^{-M q_n}, \qquad R_n(T) = e^{\varepsilon q_n} \Big[1 + \min\Big(\frac{T}{\lambda^{q_n}}, \frac{1}{\lambda^{q_n}|\delta_n|}\Big)\Big].$$

Proof. Write $q = q_n$, $A = \lambda^q$, $\delta = \delta_n$, $b = b_n$, and note $A|\delta| = e^{(L-b)q} \le e^{-\eta q}$.

The barrier radius. Apply Proposition 4.4 with $c = L + \eta$ and $\varepsilon' = \min(\varepsilon/2, L/8)$. Since $(b-L)_+ = b - L$ and $e^{(b-L)q} = 1/(A|\delta|)$, it gives $N_n \le 2e^{\varepsilon' q}/(A|\delta|) \le e^{\varepsilon q}/(A|\delta|) =: R_a$ for large q, and

$$P(R_a, T) \le P(N_n, T) \le C(1+T^2)e^{-\varepsilon' N_n/900} \le Ce^{2\Lambda q}\exp(-\varepsilon' e^{\varepsilon' q/2}/900) \le e^{-Mq}.$$

The filtered evolution. Fix $\zeta = \varepsilon/8$ and integers $k > M/L+1$ and $r > M/\varepsilon$. Choose $h \in (L, L+\eta)$ so close to L that the bounds of Lemma 5.2 and Proposition 5.3 for orders up to $r+2$ are at most $e^{\zeta q}$ after absorbing powers of q; this is possible since they are polynomial in q times powers of $E_h = e^{(h-L)q}$ of bounded order. Since $h < L + \eta$, Proposition 3.3 gives $\|R\| \le C_h q^{5/2} e^{(h-L-\eta)q} \to 0$, so $\|K\| \le 3$ for large q.

By Lemma 5.2(1), $f(B) = \sum_b g_b(H)$, so $\delta_0 = \sum_b g_b(H)\delta_0 + (1 - f(B))\delta_0$.

The remainder. Since $0 \le 1 - f \le 1$ and $1 - f(y) = 0$ for $|y - y_0| < \rho$, the spectral theorem gives $\|(1-f(B))\delta_0\| \le \rho^{-k}\|(B-y_0)^k\delta_0\|$. If v is supported in $[-jq, jq]$ with $j < k$, then $(B-y_0)v$ is supported in $[-(j+1)q, (j+1)q]$, and on that set $|V(i)/A - y_0| \le 4\pi kq|\delta|$. Hence $\|(B-y_0)v\| \le (4\pi kq|\delta| + 3/A)\|v\|$. For large q, $q|\delta| \le qe^{-(L+\eta)q} \le 1/A$, so

$$\|(1-f(B))\delta_0\| \le \Big(\frac{C_k}{\rho A}\Big)^k \le \big(16C_k q^4 e^{-Lq}\big)^k \le e^{-Mq}.$$

This vector evolves unitarily, so its contribution to $P(\cdot, T)$ is at most $2e^{-2Mq}$ at every radius.

The retained part. By Proposition 5.3(1), $\sum_b e^{-itH} g_b(H)\delta_0 = \sum_b e^{-itH_b} g_b(H)\delta_0 =: \Phi_t$. By Lemma 5.2 and Proposition 5.3, each branch satisfies the hypotheses of Proposition 5.3(4) with $M_0 \le e^{\zeta q}$ and $\gamma \le e^{2\zeta q}/A$. Summing over q branches,

$$\|X^r \Phi_t\| \le C_r q e^{\zeta q}(1 + |t|e^{2\zeta q}/A)^r.$$

Let $R_b = e^{\varepsilon q}(1 + T/A)$. By Markov's inequality $\|\mathbf{1}_{|X|>R_b}\Phi_t\| \le R_b^{-r}\|X^r\Phi_t\|$. Averaging against $(2/T)e^{-2t/T}$, whose moments satisfy $\int t^j (2/T)e^{-2t/T}dt \le C_j T^j$,

$$\frac{2}{T}\int_0^\infty e^{-2t/T}\|\mathbf{1}_{|X|>R_b}\Phi_t\|^2\,dt \le C\,q^2 e^{2\zeta q}\,\frac{(1+Te^{2\zeta q}/A)^{2r}}{R_b^{2r}} \le Cq^2 e^{(2\zeta + 4r\zeta - 2r\varepsilon)q} \le e^{-Mq}.$$

Combining the two parts with $\|x+y\|^2 \le 2\|x\|^2 + 2\|y\|^2$ gives $P(R_b, T) \le Ce^{-Mq}$.

Conclusion. $R_n(T) \ge \min(R_a, R_b)$, and $P(\cdot, T)$ is nonincreasing, so $P(R_n(T), T) \le \max(P(R_a, T), P(R_b, T)) \le Ce^{-Mq}$. Every constant was fixed before n, θ and T vary. □

The radius $R_n(T)$ describes the motion at the scale q_n. Up to the factor $e^{\varepsilon q}$, the packet moves with speed λ^{-q} until it reaches the barrier radius $1/(\lambda^q|\delta_n|)$, where the drift of the diagonal stops it.

6 A per-denominator estimate

The barrier and slow-dynamics estimates need a definite Liouville ratio at the scale used. Times governed by scales with small ratios are handled by the complex-energy method of Damanik and Tcheremchantsev. We use it only in the following form.

Proposition 6.1. *There are constants $C_D \ge 1$ and $C, c_D, \kappa > 0$ such that for every phase θ, every sufficiently large n, every $T \ge 1$ and every $N \ge C_D q_n$,*

$$P(N, T) \le Ce^{-c_D N} + CT^3 e^{-\kappa q_n}.$$

No assumption on q_{n+1} or on $\beta(\alpha)$ enters. The estimate confines the packet to distance of order q_n up to times $e^{\kappa q_n/3}$.

The proof combines two ingredients. The first is the general transport bound of Damanik and Tcheremchantsev [11, Theorem 7]. For a Schrödinger operator with bounded potential, choose $K_E \geq 4$ with $\sigma(H) \subset [-K_E+1, K_E-1]$; here $K_E = \max(4, 2\lambda+3)$ works, since $\|H\| \leq 2+2\lambda$. Their theorem bounds the right outside probability, for the initial state δ_1 and the same exponential time average, by

$$C\exp(-cN) + CT^3 \int_{-K_E}^{K_E} \Big(\max_{1\leq k\leq N} \|\Phi(k, E+i/T)\|^2 \Big)^{-1} dE,$$

where $\Phi(k,z)$ is the transfer matrix from site 1 to site k, and the left outside probability by the analogous expression for negative k. The constants depend only on K_E and the bound on the potential. Changing the initial site from δ_1 to δ_0 shifts the phase by α and the radius by one, which is absorbed by enlarging C_D and C.

The second ingredient is a uniform lower bound for transfer matrices at complex energies. It follows the argument of [11, Section 5], with a slightly smaller auxiliary block, which gives an orbit-hitting step with room to spare.

Lemma 6.2. *There is $\gamma > 0$ such that for all sufficiently large n, every phase x and every z with $|\operatorname{Re} z| \leq K_E$ and $0 \leq \operatorname{Im} z \leq 1$,*

$$\max_{1\leq k\leq 2q_n} \|A_k(z,x)\| \geq e^{\gamma q_n}.$$

Proof. Let $\Gamma > L$ satisfy $\|A_r(z,x)\|_{\mathrm{HS}} \leq e^{\Gamma r}$ for all $r \geq 1$, $x \in \mathbb{T}$ and z in the strip; this is possible because the one-step matrices are uniformly bounded there.

Herman's bound. With $w = e^{2\pi i x}$, the entry $(A_r(z,x))_{11}$ is a Laurent polynomial in w. Its term of lowest degree, w^{-r}, comes only from the product of the diagonal entries and has coefficient $\prod_{k<r}(-\lambda e^{-2\pi i k\alpha})$, of modulus λ^r. So $p(w) = w^r (A_r(z,x))_{11}$ is a polynomial with $|p(0)| = \lambda^r$, and Jensen's formula gives

$$\int_{\mathbb{T}} \log \|A_r(z,x)\|\, dx \geq \int_{\mathbb{T}} \log |p(e^{2\pi i x})|\, dx \geq rL.$$

A long good interval. Let $\rho_0 = L/(2\Gamma - L)$ and $G_r = \{x : \|A_r(z,x)\|_{\mathrm{HS}} > e^{rL/2}\}$. Splitting the integral over G_r and its complement gives $rL \leq |G_r|\Gamma r + (1-|G_r|)rL/2$, so $|G_r| \geq \rho_0$. The function $\|A_r(z,x)\|_{\mathrm{HS}}^2$ is a real trigonometric polynomial of degree at most $2r$ in x, so unless $G_r = \mathbb{T}$ it has at most $4r$ components, and one of them is an interval of length at least $\rho_0/(4r)$.

Orbit hitting. Let $q = q_n$ and $r = \lfloor \rho_0 q/32 \rfloor$. Then that interval has length at least $8/q$. The points $x + j\alpha$, $0 \leq j < q$, are within $1/q$ of a translate of the grid $q^{-1}\mathbb{Z}$, because $|j\alpha - jp_n/q| < j/q^2 < 1/q$. Hence they are $3/(2q)$-dense, and some $j < q$ has $x + j\alpha \in G_r$, so $\|A_r(z, x+j\alpha)\| \geq 2^{-1/2} e^{rL/2}$.

Conclusion. Since $A_{j+r}(z,x) = A_r(z, x+j\alpha) A_j(z,x)$ and $\|M^{-1}\| = \|M\|$ for a 2×2 matrix of determinant one,

$$2^{-1/2} e^{rL/2} \leq \|A_{j+r}(z,x)\|\, \|A_j(z,x)\|.$$

So one of $\|A_j\|$ (if $j \geq 1$) or $\|A_{j+r}\|$ is at least $2^{-1/4} e^{rL/4}$, with $1 \leq j + r \leq 2q$. Since $r \geq \rho_0 q/32 - 1$, this proves the lemma with any $\gamma < \rho_0 L/128$. □

Proof of Proposition 6.1. For $T \geq 1$ and $|E| \leq K_E$, the point $E + i/T$ lies in the strip of Lemma 6.2. For the right half-line, the transfer matrices from site 1 are $A_k(z, \theta+\alpha)$, so the integrand in [11, Theorem 7] is at most $e^{-2\gamma q_n}$ once $N \geq 2q_n$. For the left half-line, reflection $n \mapsto -n$ maps $H_{\lambda,\alpha,\theta}$

to $H_{\lambda,\alpha,-\theta}$, because the cosine is even; so the left transfer matrices are right transfer matrices for another phase, and the same bound holds. With the matrix conventions of [11] one can equally use the identity $\Phi(n,z,\alpha,\theta) = R\,\Phi(-n,z,-\alpha,\alpha+\theta)\,R$ with $R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which preserves norms. Integration over $|E| \le K_E$ gives the claim with $\kappa = 2\gamma$ and $C_D = 3$, the extra unit absorbing the change of initial site. Finitely many small n are absorbed into C, since $P \le 1 \le T^3$. □

7 The upper bound

In this section $\beta = \beta(\alpha) < \infty$ and $\theta \in G_\alpha$ is fixed. Let $n_G(\theta)$ be an index such that $|\sin(2\pi q_n\theta)| \ge q_n^{-2}$ for all $n \ge n_G(\theta)$. The thresholds n_0 below may depend on θ through $n_G(\theta)$, and the time threshold in Lemma 7.2 may depend on θ as well. The constants C do not, since Proposition 4.4 and Theorem 5.4 are uniform over phases satisfying the sine condition at the scale used. All of these are fixed before n and T vary.

7.1 Logarithmic windows at scales with a definite ratio

Proposition 7.1. *Let $0 < a < 1$, $m > 0$, $c > 0$ and $0 < d < B$, and let $B_0 > \beta$ satisfy $f_0 := \max\{0, 1 - L/B_0\} < a$. There are C and n_0 such that for every $n \ge n_0$ with $b_n \ge c$ and every T with $e^{dq_n} \le T \le e^{Bq_n}$,*

$$P(T^a, T) \le CT^{-m}.$$

Proof. Choose $\eta > 0$ with $2\eta < ad/4$, then $\varepsilon > 0$ with $\varepsilon < ad/4$ and $\varepsilon/d < (a - f_0)/2$. Since $\limsup b_n = \beta < B_0$, we may take $n_0 \ge n_G(\theta)$ so large that $b_n \le B_0$ for $n \ge n_0$. Write $q = q_n$, $b = b_n$ and $\tau = \log T/q \in [d, B]$.

Case $b \ge L + 2\eta$. Apply Theorem 5.4 with this η and ε, $\Lambda = \max(1, B)$ and $M = mB$. It gives $P(R_n(T), T) \le Ce^{-mBq} \le CT^{-m}$. Moreover

$$\frac{\log R_n(T)}{\log T} \le \frac{\varepsilon}{d} + \frac{\log 2}{dq} + \frac{\max\{0, \min(\tau - L,\, b - L)\}}{\tau}.$$

The last term is at most $1 - L/b \le f_0$, since $b > L$. Indeed, it equals 0 if $\tau \le L$, equals $1 - L/\tau \le 1 - L/b$ if $L < \tau \le b$, and equals $(b - L)/\tau \le (b - L)/b$ if $\tau \ge b$. Hence $\log R_n(T)/\log T < (a + f_0)/2 + o(1) < a$ for large n, so $R_n(T) \le T^a$ and $P(T^a, T) \le P(R_n(T), T) \le CT^{-m}$.

Case $c \le b < L + 2\eta$. Apply Proposition 4.4 with $\varepsilon_2 = \min\{\varepsilon, \min(L, c)/8\}$. It gives $N_n \le 2e^{(2\eta+\varepsilon_2)q} \le 2e^{adq/2} \le T^a$ for large q, since $2\eta + \varepsilon_2 < ad/2$. Hence

$$P(T^a, T) \le P(N_n, T) \le C(1 + e^{2Bq}) \exp(-\varepsilon_2 e^{\varepsilon_2 q/2}/900) \le e^{-mBq} \le T^{-m}$$

for large q. All parameters were fixed before n and T vary. □

When $\beta \le L$ one may take $B_0 < L + 2\eta$, and then only the second case occurs.

7.2 Covering every time

Lemma 7.2. *Let $a \in (0, 1)$ with $a > \max\{0, 1 - L/\beta\}$ and let $m > 0$. Then $P(T^a, T) = O(T^{-m})$ as $T \to \infty$.*

Proof. Let C_D, κ be as in Proposition 6.1. Fix $K \geq 1$ with $\kappa K > m+3$, and put $d = 1/K$ and $c = a/(2K)$. Fix $B_0 > \beta$ with $\max\{0, 1 - L/B_0\} < a$, and put $B = \max\{2B_0/a, 2d\}$. Let C and n_0 be given by Proposition 7.1 for these parameters and the fixed phase θ.

For large T let j be the least index with $q_j \geq K \log T$. Two cases exhaust all possibilities.

Case A: $q_j \leq T^a/(2C_D)$. Let $N = \lfloor T^a \rfloor$, so $N \geq C_D q_j$ for large T. Proposition 6.1 gives

$$P(T^a, T) \leq P(N, T) \leq Ce^{-c_D(T^a-1)} + CT^3 e^{-\kappa q_j} \leq Ce^{-c_D(T^a-1)} + CT^{3-\kappa K} = O(T^{-m}).$$

This case uses no information about the phase and allows arbitrarily long stretches of scales with small ratios.

Case B: $q_j > T^a/(2C_D)$. Let $q = q_{j-1}$. By the choice of j, $q < K \log T$, so $\tau = \log T/q > d$. By (6), $q_j < D_{j-1} = e^{b_{j-1}q}$, so

$$a \log T - \log(2C_D) < \log q_j < b_{j-1} q.$$

Along a sequence of times in this case, $q_j \to \infty$, hence $j \to \infty$ and $q \to \infty$. Dividing by $q < K \log T$ gives $b_{j-1} > a/K - \log(2C_D)/q \geq a/(2K) = c$ for large T. Also $b_{j-1} \leq B_0$ for large j, so $\tau < (B_0 + \log(2C_D)/q)/a \leq B$. Thus $T \in [e^{dq}, e^{Bq}]$ and $b_{j-1} \geq c$. Since $\theta \in G_\alpha$ and $j - 1 \geq n_0$ for large T, Proposition 7.1 gives $P(T^a, T) \leq CT^{-m}$. □

The covering needs no information about the scales between q_{j-1} and q_j other than the two inequalities above. In particular it allows $\liminf_n \beta_n = 0$.

Theorem 7.3. *Let $\lambda > 1$, let α be irrational with $\beta(\alpha) < \infty$, and let $\theta \in G_\alpha$. For every $a > \max\{0, 1 - L/\beta(\alpha)\}$ and every $m > 0$, $P(T^a, T) = O(T^{-m})$. Consequently*

$$\alpha_u^+(H_{\lambda,\alpha,\theta}) \leq \max\Big\{0,\ 1 - \frac{\log \lambda}{\beta(\alpha)}\Big\}.$$

Proof. For $a < 1$ this is Lemma 7.2. For $a \geq 1$ choose $a' \in (\max\{0, 1 - L/\beta\}, 1)$ and use $P(T^a, T) \leq P(T^{a'}, T)$. The bound on α_u^+ follows from Lemma 2.2(3), letting a decrease to $\max\{0, 1 - L/\beta\}$. □

8 A lower bound for Lipschitz potentials

This section proves a lower bound for the fastest exponent α_u^+. It holds for every phase and for general Lipschitz sampling functions. For the almost Mathieu operator, Section 9 strengthens it to every moment. In this section only, let $v : \mathbb{T} \to \mathbb{R}$ be Lipschitz, let $H_\theta u(n) = u(n+1) + u(n-1) + v(\theta + n\alpha)u(n)$, and let $T(E, x)$, $A_r(E, x)$, $L(E)$ and Σ be defined as in Section 2 with this v.

Theorem 8.1. *Let α be irrational with $0 < \beta(\alpha) < \infty$. Assume that $L(\cdot)$ is continuous on a neighborhood of a point $E_* \in \Sigma$ with $L(E_*) = \ell := \min_\Sigma L$. Then for every phase θ,*

$$\alpha_u^+(H_\theta) \geq \max\Big\{0,\ 1 - \frac{\ell}{\beta(\alpha)}\Big\}.$$

For the almost Mathieu operator with $\lambda > 1$, $L(E)$ is continuous in E and equals $\log \lambda$ on Σ [6, Theorem 1 and Corollary 2], so $\ell = \log \lambda$ and $\alpha_u^+ \geq 1 - \log \lambda/\beta(\alpha)$ for every phase.

8.1 Elliptic blocks

For a finite sequence of energies $E - v_r$, $0 \le r < q$, let $T_r = \begin{pmatrix} E - v_r & -1 \\ 1 & 0 \end{pmatrix}$, $C_0 = I$, $C_r = T_{r-1} \cdots T_0$ and $\mathcal{A} = C_q$. Suppose $\mathcal{A}$ is real and elliptic, $|\operatorname{tr} \mathcal{A}| < 2$. Choose $B \in SL(2, \mathbb{R})$ with $B^{-1} \mathcal{A} B = R_\varrho$, the rotation by angle ϱ, and put $u_r = e_1^{\mathsf{T}} C_r B$, the first row of $C_r B$, and

$$S_{\mathcal{A}} = \sum_{r=0}^{q-1} \|u_r\|^2, \qquad N_\kappa = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Lemma 8.2. *1. $S_{\mathcal{A}} = |\partial_E \operatorname{tr} \mathcal{A}| / |\sin \varrho|$.*

2. For every $z \in \mathbb{C}$ and $0 \le r \le q$, with $\mathcal{A}(E+z)$ and $C_r(E+z)$ the products at the energy $E+z$,

$$\|B^{-1} \mathcal{A}(E+z) B\| \le e^{|z| S_{\mathcal{A}}}, \qquad \|C_r(E+z) B\| \le \|C_r(E) B\| \, e^{|z| S_{\mathcal{A}}}.$$

3. If $|\operatorname{tr} \mathcal{A}| \le 3/2$ and the entries v_r depend on a parameter ϕ with $|\partial_\phi v_r| \le \operatorname{Lip}$, then $\|B^{-1}(\partial_\phi \mathcal{A}) B\| \le C \operatorname{Lip} |\partial_E \operatorname{tr} \mathcal{A}|$ and $\|B^{-1}(\partial_E \mathcal{A}) B\| \le C |\partial_E \operatorname{tr} \mathcal{A}|$, with an absolute constant C.

4. Let $z_{\mathcal{A}}$ be the fixed point of $\mathcal{A}$ in the upper half-plane. Any $B \in SL(2, \mathbb{R})$ with $B(i) = z_{\mathcal{A}}$ conjugates $\mathcal{A}$ to a rotation, and $\|B\|^2 = \|B^{-1}\|^2 = e^{d(i, z_{\mathcal{A}})}$, where d is the hyperbolic distance. If $|\operatorname{tr} \mathcal{A}| \le 3/2$, then $e^{d(i, z_{\mathcal{A}})} \le C \|\mathcal{A}\|$, and $z_{\mathcal{A}}$ moves with hyperbolic speed at most $C \|B^{-1} \mathcal{A}' B\|$ under any parameter derivative $\mathcal{A}'$.

Proof. Put $Q_r = B^{-1} C_r^{-1} N_\kappa C_r B$. Since $T_r(E+z) = T_r(E)(I + z N_\kappa)$, telescoping gives

$$B^{-1} \mathcal{A}(E+z) B = R_\varrho \prod_{r=q-1}^{0} (I + z Q_r),$$

with the factors ordered from $r = q-1$ on the left to $r = 0$ on the right. A direct computation gives $Q_r = -J_0 u_r^{\mathsf{T}} u_r$, a rank-one matrix with $\|Q_r\| = \|u_r\|^2$. Hence $\|B^{-1} \mathcal{A}(E+z) B\| \le \prod_r (1 + |z| \|u_r\|^2) \le e^{|z| S_{\mathcal{A}}}$, and the same computation truncated at r gives the second inequality in part 2. Differentiating at $z = 0$, $B^{-1}(\partial_E \mathcal{A}) B = R_\varrho \sum_r Q_r$, and the trace of $R_\varrho(-J_0 u_r^{\mathsf{T}} u_r)$ equals $\sin \varrho \, \|u_r\|^2$, because the symmetric part of $R_\varrho J_0$ is $-\sin \varrho \, I$ and the antisymmetric part has zero quadratic form. This proves part 1. A change of v_r at rate c_r gives in the same way $B^{-1} \mathcal{A}' B = R_\varrho \sum_r c_r Q_r$, so $\|B^{-1} \mathcal{A}' B\| \le \max_r |c_r| \, S_{\mathcal{A}}$. When $|\operatorname{tr} \mathcal{A}| \le 3/2$, $|\sin \varrho| \ge \sqrt{7}/4$, and part 3 follows from part 1. For part 4, recall that every $M \in SL(2, \mathbb{R})$ satisfies $\|M\|_{\mathrm{HS}}^2 = 2 \cosh d(i, M(i))$; since its singular values are σ and σ^{-1}, this gives $\|M\|^2 = e^{d(i, M(i))}$. Apply this to B and B^{-1}. If $B(i) = z_{\mathcal{A}}$, then $B^{-1} \mathcal{A} B$ fixes i and is a rotation. Conjugating to that rotation and differentiating $\mathcal{A}(z_{\mathcal{A}}) = z_{\mathcal{A}}$ shows that the fixed point moves at hyperbolic speed at most $|1 - e^{-2i\varrho}|^{-1} \|B^{-1} \mathcal{A}' B\|$. Finally $\|\mathcal{A}\|_{\mathrm{HS}}^2 = 2 + 4 \sinh^2\big(d(i, z_{\mathcal{A}})\big) \sin^2 \varrho$, which with $|\sin \varrho| \ge \sqrt{7}/4$ gives $e^{d(i, z_{\mathcal{A}})} \le C \|\mathcal{A}\|$. □

8.2 Proof of Theorem 8.1

If $\beta(\alpha) \le \ell$ there is nothing to prove, so assume $\ell < \beta(\alpha)$. Fix h with $\ell < h < \beta(\alpha)$.

Uniform real growth. By continuity of L near E_* there is a compact neighborhood U of E_* on which $L < h$. The argument of Lemma 2.3 gives C_h with $\|A_r(E, x)\| \le C_h e^{hr}$ for $E \in U$, $x \in \mathbb{T}$ and $r \ge 0$. By the product rule, with $D_q = C q e^{hq}$,

$$|\partial_E \operatorname{tr} A_q(E, x)| \le D_q, \qquad |\partial_x \operatorname{tr} A_q(E, x)| \le \operatorname{Lip}(v) \, D_q \qquad (E \in U), \tag{11}$$

the second almost everywhere in x.

A band near E_*. Choose convergents with $\beta_n \to \beta(\alpha)$; write $q = q_n$, $\delta = \delta_n$. Fix a phase x_0 and let H^{per} be the q-periodic operator whose potential repeats $v(x_0 + k\alpha)$, $0 \le k < q$. Its monodromy is $A_q(E, x_0)$, and it differs in norm by at most $\mathrm{Lip}(v)|\delta|$ from the p/q-periodic operator at phase x_0. The proof of Lemma 2.5, which uses only the Lipschitz bound on v, shows that the phase-union spectrum of the p/q-periodic operators converges to Σ and lies within $\mathrm{Lip}(v)/(2q)$ of each fixed-phase spectrum. So $\sigma(H^{\mathrm{per}})$ has a point converging to E_*. Every band of a q-periodic operator with hopping one has width at most $2\pi/q$: distributing the Bloch phase over the q bonds makes the Floquet matrices $2/q$-Lipschitz in the Bloch phase. Hence, for large n, an entire band of H^{per} lies in U. On that band $\operatorname{tr} A_q(E, x_0)$ is monotone and takes all values in $[-2, 2]$, so the set

$$S_q = \{E \text{ in the band} : |\operatorname{tr} A_q(E, x_0)| \le 1\}$$

is an interval of length at least $2/D_q$.

Slowly drifting blocks. The blocks of the actual operator starting at sites mq, for $\theta = x_0$, are $A^{(m)}(E) = A_q(E, x_0 + mq\alpha) = A_q(E, x_0 + m\delta)$, since $q\alpha = p + \delta$. Let

$$K_q = \Big\lfloor \frac{c_0}{D_q|\delta|} \Big\rfloor, \qquad N_q = qK_q, \qquad T_q = C_0 K_q D_q,$$

with c_0 small and C_0 large fixed constants. Since $\log(1/|\delta|) = b_n q$ and $b_n \to \beta(\alpha) > h$, $K_q \to \infty$. For $E \in S_q$ and $0 \le m < K_q$, (11) gives $|\operatorname{tr} A^{(m)}(E) - \operatorname{tr} A_q(E, x_0)| \le \mathrm{Lip}(v) D_q K_q |\delta| \le \mathrm{Lip}(v) c_0 \le 1/2$, so every block satisfies $|\operatorname{tr} A^{(m)}| \le 3/2$.

By Lemma 8.2(3)–(4), the fixed points z_m of the blocks $A^{(m)}$ move a total hyperbolic distance $\sum_m d(z_m, z_{m+1}) \le C\,\mathrm{Lip}(v) D_q K_q |\delta| \le C$. Choose any $B_m \in SL(2, \mathbb{R})$ with $B_m(i) = z_m$. Since $B_{m+1}^{-1} B_m$ maps i to a point at distance $d(z_m, z_{m+1})$ from i, Lemma 8.2(4) gives $\|B_{m+1}^{-1} B_m\| = e^{d(z_m, z_{m+1})/2}$, so $\prod_m \|B_{m+1}^{-1} B_m\| \le e^C$. By Lemma 8.2(1)–(2), each block at the energy $E + i/T_q$ satisfies $\|B_m^{-1} A^{(m)}(E + i/T_q) B_m\| \le e^{CD_q/T_q}$, and the product of K_q such factors is at most $e^{CK_q D_q/T_q} \le e^{C/C_0}$. The endpoint lifts cost at most Ce^{hq} by Lemma 8.2(4). A partial last block is handled by the second inequality in Lemma 8.2(2). Hence, writing $\Phi(x, z)$ for the transfer matrix of H_{x_0} from site 0 to site x,

$$\max_{0 \le x \le N_q} \|\Phi(x, E + i/T_q)\| \le Ce^{2hq} \qquad (E \in S_q). \tag{12}$$

Resolvent. Let $G(n; z) = \langle \delta_n, (H_{x_0} - z)^{-1} \delta_0 \rangle$. For $n \ge 1$ the vector $(G(n+1; z), G(n; z))$ is obtained from $(G(1; z), G(0; z))$ by the transfer matrices starting at site 1, which are bounded as in (12) after a shift of the starting site that changes only constants. Since these matrices have determinant one and $|G(0; E + i/T)| \ge \operatorname{Im} G(0; E + i/T) \ge c/T$ for bounded E,

$$|G(n; E + i/T_q)|^2 + |G(n+1; E + i/T_q)|^2 \ge cT_q^{-2} e^{-4hq} \qquad (E \in S_q,\ 0 \le n \le N_q).$$

By Parseval's formula, $a(n, T) = (\pi T)^{-1} \int_{\mathbb{R}} |G(n; E + i/T)|^2\, dE$. Summing over the pairs of sites between $N_q/2$ and N_q gives

$$P(N_q/2, T_q) \ge c\, N_q\, |S_q|\, T_q^{-3} e^{-4hq} \ge T_q^{-C_*} \tag{13}$$

for a finite constant C_* depending only on h and $\beta(\alpha)$, because $\log T_q = b_n q + O(\log q)$ and $|S_q| \ge 2/D_q$.

The exponent. From $\log N_q = (b_n - h)q + O(\log q)$ and $\log T_q = b_n q + O(\log q)$ we get $\log N_q / \log T_q \to 1 - h/\beta(\alpha)$. By (13), $M_p(T_q) \ge (N_q/2)^p P(N_q/2, T_q) \ge c(N_q/2)^p T_q^{-C_*}$, so

$\beta^+(p) \geq 1 - h/\beta(\alpha) - C_*/p$. Letting $p \to \infty$ and then $h \downarrow \ell$ proves the theorem for the phase x_0. The phase was arbitrary. □

The mass in (13) is only polynomially small, so this argument bounds α_u^+ but not each fixed $\beta^+(p)$. The losses are the pointwise bound $|G(0)| \geq c/T$, the two frame changes at the ends of the run, and an energy window of small spectral weight. Section 9 removes all three for the almost Mathieu operator.

9 Every moment and the exact packing dimension

The lower bound of Section 8 produces only a polynomially small mass at the critical distance. In this section we show that for the almost Mathieu operator a fixed fraction of the packet gets there. Two facts drive the proof. Under the spectral measure, the rational discriminant at the phase θ is asymptotically arcsine distributed, so most of the spectral mass sits at energies whose first q-blocks are elliptic with a definite margin (Lemma 9.1). At such an energy, slightly above the real axis, the Weyl solution stays spread over a long run of consecutive q-blocks (Lemma 9.5). The same runs bound the spectral mass of small intervals, which gives the packing dimension. The dependencies are: Lemmas 9.1 and 9.5 give Theorem 9.7, which gives Corollary 9.9 and Theorems A and B; Lemma 9.5 gives Lemma 9.10 and part 1 of Theorem C; Theorem A gives Proposition 9.11 and part 2 of Theorem C.

Throughout, $\lambda > 1$, $v(x) = 2\lambda \cos 2\pi x$ and $L = \log \lambda$, except in Subsection 9.2, where v is any Lipschitz function. For a phase θ we write μ^θ for the spectral measure of δ_0 for $H_{\lambda,\alpha,\theta}$, and

$$G(k; z) = \langle \delta_k, (H_{\lambda,\alpha,\theta} - z)^{-1} \delta_0 \rangle \qquad (k \in \mathbb{Z},\ \operatorname{Im} z > 0).$$

Two identities connect these objects with transport. The first is Parseval's formula, already used in Section 8:

$$a(k, T) = \frac{1}{\pi T} \int_{\mathbb{R}} |G(k; E + i/T)|^2 \, dE. \tag{14}$$

It follows from $(H - E - i/T)^{-1} = i \int_0^\infty e^{-t/T} e^{itE} e^{-itH} \, dt$ and Plancherel's theorem. The second is the spectral theorem: for $z = E + i\varepsilon$ with $\varepsilon > 0$,

$$\sum_{k \in \mathbb{Z}} |G(k; z)|^2 = \frac{\operatorname{Im} G(0; z)}{\varepsilon}, \qquad \frac{1}{\pi} \operatorname{Im} G(0; z) = \int_{\mathbb{R}} P_\varepsilon(E - E') \, d\mu^\theta(E'), \qquad P_\varepsilon(x) = \frac{\varepsilon}{\pi(x^2 + \varepsilon^2)}. \tag{15}$$

9.1 The arcsine law

Fix a scale n and write $q = q_n$, $p = p_n$, $\delta = \delta_n$. For the phase θ put

$$\Delta_n(E) = P_q(E) - 2\lambda^q \cos(2\pi q \theta),$$

the discriminant of the periodic operator $H_{\lambda,p/q,\theta}$ (Lemma 3.1). For that periodic operator the distribution of Δ_n under the spectral measure of δ_0 is exactly the arcsine law: by (7), $\Delta_n(H_{\lambda,p/q,\theta}) = S^q + S^{-q}$. The next lemma transfers this to the irrational operator at every scale where $|\delta_n|$ is much smaller than λ^{-q_n}.

Let σ be the arcsine law on $[-2, 2]$, with density $1/(\pi\sqrt{4 - x^2})$. Its moments are $m_k = \binom{k}{k/2}$ for even k and $m_k = 0$ for odd k, and $\sigma([-a, a]) = \frac{2}{\pi} \arcsin(a/2)$ for $0 \leq a \leq 2$.

Lemma 9.1 (Arcsine law)**.** *Let $c > L$ and let $\mathcal{N}$ be an infinite set of indices with $b_n \geq c$ for $n \in \mathcal{N}$. For every phase θ let ν_n be the image of μ^θ under Δ_n. Then $\int x^k \, d\nu_n \to m_k$ for every integer $k \geq 0$, uniformly in θ, as $n \to \infty$ in $\mathcal{N}$. Consequently ν_n converges weakly to σ, and for every $0 \leq \eta < 2$,*

$$\lim_{n \in \mathcal{N}} \mu^\theta(\{E : |\Delta_n(E)| \leq 2 - \eta\}) = \frac{2}{\pi} \arcsin\Big(1 - \frac{\eta}{2}\Big).$$

Proof. Fix $h \in (L, c)$. By Proposition 3.3, for large n and every phase,

$$\Delta_n(H) = Q - V(0) = D + F + R, \qquad D = V - V(0), \quad F = S^q + S^{-q},$$

where D is diagonal and R has range at most q and norm at most $C_h q^{5/2} e^{hq} |\delta|$. Each of D, F and R maps a vector supported in $[-jq, jq]$ to a vector supported in $[-(j+1)q, (j+1)q]$.

Fix $k \geq 1$. In the expansion of $\Delta_n(H)^k \delta_0$ the operator D only acts on vectors supported in $[-kq, kq]$, where

$$|D(i)| = 2\lambda^q |\cos(2\pi(q\theta + i\delta)) - \cos(2\pi q\theta)| \leq 4\pi \lambda^q \, kq |\delta|.$$

So $\Delta_n(H)^k \delta_0 = (D_k + F + R)^k \delta_0$ with $D_k = D\mathbf{1}_{[-kq,kq]}$. Put $\varepsilon_n = 4\pi kq\lambda^q |\delta| + C_h q^{5/2} e^{hq} |\delta|$, so that $\|D_k + R\| \leq \varepsilon_n$, and recall $\|F\| \leq 2$. Expanding the product,

$$\big\|\Delta_n(H)^k \delta_0 - F^k \delta_0\big\| \leq (2 + \varepsilon_n)^k - 2^k.$$

Since $b_n \geq c > h > L$, both $q\lambda^q |\delta| = q e^{(L - b_n)q}$ and $q^{5/2} e^{hq} |\delta| = q^{5/2} e^{(h - b_n)q}$ tend to 0, uniformly in θ, so $\varepsilon_n \to 0$. Next, $F^k \delta_0 = \sum_i \binom{k}{(k+i)/2} \delta_{iq}$, the sum running over $|i| \leq k$ with $i \equiv k \pmod 2$, so $\langle \delta_0, F^k \delta_0 \rangle = m_k$. Since $\int x^k \, d\nu_n = \langle \delta_0, \Delta_n(H)^k \delta_0 \rangle$, we obtain $|\int x^k \, d\nu_n - m_k| \leq (2 + \varepsilon_n)^k - 2^k \to 0$.

The law σ has compact support, so it is determined by its moments [5, Theorem 30.1], and convergence of all moments implies $\nu_n \Rightarrow \sigma$ [5, Theorem 30.2]. Since σ has no atoms, weak convergence gives $\nu_n(J) \to \sigma(J)$ for every interval J. Take $J = [-2 + \eta, 2 - \eta]$. □

The proof uses no condition on the phase. The only arithmetic input is $b_n \geq c > L$, which makes the residual ε_n small on the sites that k steps from the origin can reach.

9.2 One elliptic run

In this subsection $v : \mathbb{T} \to \mathbb{R}$ is Lipschitz with constant $\mathrm{Lip}(v)$, and $T(E, x)$ and $A_r(E, x)$ are defined with this v as in Section 8. Fix a phase θ and a convergent p/q of α, and put $\delta = q\alpha - p$, so that $x + q\alpha = x + \delta$ in $\mathbb{T}$. Let $z = E + i\varepsilon$ with $E \in \mathbb{R}$ and $\varepsilon > 0$, and let $g \in \ell^2(\mathbb{Z}_{\geq 0})$ satisfy

$$g(k+1) + g(k-1) + v(\theta + k\alpha)\, g(k) = z\, g(k) \qquad (k \geq 1). \tag{16}$$

Cut the half-line into blocks of q sites. Put $x_m = \theta + (mq + 1)\alpha$, $\Psi_m = (g(mq + 1), g(mq))^{\mathsf{T}}$ and

$$\mathcal{M}_m = \sum_{r=0}^{q-1} |g(mq + 1 + r)|^2.$$

Then $(g(mq + 1 + r), g(mq + r))^{\mathsf{T}} = A_r(z, x_m)\Psi_m$ for $0 \leq r \leq q$, and in particular $\Psi_{m+1} = A_q(z, x_m)\Psi_m$. Consecutive blocks see almost the same potential: $x_{m+1} = x_m + \delta$, so the potentials at corresponding sites differ by at most $\mathrm{Lip}(v)|\delta|$. We use the notation T_r, C_r, N_κ, J_0 and R_ϱ of Lemma 8.2, and d for the hyperbolic distance in the upper half-plane, normalized as there.

The first lemma is the telescoping identity behind Lemma 8.2, for site-dependent shifts.

Lemma 9.2 (Insertion). *Let T_r and C_r be as in Lemma 8.2 for energies $E - v_r$, $0 \le r < q$, let $B \in SL(2,\mathbb{R})$, and put $u_r = e_1^{\mathsf T} C_r B$. Let $w_0, \dots, w_{q-1} \in \mathbb{C}$ and let C_r' be the products for the energies $E - v_r + w_r$. Then for $0 \le r \le q$,*

$$C_r' B = C_r B\, \Pi_r, \qquad \Pi_r = (I + w_{r-1}Q_{r-1}) \cdots (I + w_0 Q_0), \qquad Q_j = -J_0 u_j^{\mathsf T} u_j,$$

with $\det \Pi_r = 1$ and $\|\Pi_r - I\| \le \exp\bigl(\sum_{j<r} |w_j|\, \|u_j\|^2\bigr) - 1$.

Proof. The new one-step matrices are $T_j(I + w_j N_\kappa)$. If $C_j' B = C_j B \Pi_j$, then

$$C_{j+1}' B = T_j(I + w_j N_\kappa) C_j B \Pi_j = C_{j+1} B\, (I + w_j B^{-1} C_j^{-1} N_\kappa C_j B)\, \Pi_j,$$

which gives the product formula by induction. The computation in the proof of Lemma 8.2, which uses only $\det B = \det C_j = 1$, gives $B^{-1} C_j^{-1} N_\kappa C_j B = -J_0 u_j^{\mathsf T} u_j$. This matrix has norm $\|u_j\|^2$, and it has trace $-u_j J_0 u_j^{\mathsf T} = 0$ and rank one, so $\det(I + w_j Q_j) = 1$. Expanding the product gives the norm bound. □

Lemma 9.3 (Perturbed rotations). *Let $|\sin \varrho| \ge s > 0$ and let $M = R_\varrho(I + Y) \in SL(2,\mathbb{R})$ with $\|Y\| \le y \le s/8$. Then M is elliptic, and its fixed point w in the upper half-plane satisfies $d(i, w) \le 12y/s$.*

Proof. The unitary matrix $\mathcal{C} = 2^{-1/2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$ acts as the Cayley map from the upper half-plane to the unit disc, sends i to 0, and is an isometry for the hyperbolic metrics. It conjugates R_ϱ to $\operatorname{diag}(e^{-i\varrho}, e^{i\varrho})$ and M to a matrix $\begin{pmatrix} a & b \\ \bar b & \bar a \end{pmatrix}$ with $|a|^2 - |b|^2 = 1$. Since $\mathcal{C}$ is unitary, $\|\mathcal{C} M \mathcal{C}^{-1} - \mathcal{C} R_\varrho \mathcal{C}^{-1}\| = \|R_\varrho Y\| \le y$, so $|b| \le y$ and $|\operatorname{Im} a| \ge s - y \ge \frac{7}{8} s > |b|$. Hence M is elliptic. The fixed point of $\zeta \mapsto (a\zeta + b)/(\bar b \zeta + \bar a)$ in the disc solves $\bar b \zeta^2 - 2i(\operatorname{Im} a)\zeta - b = 0$ and has modulus

$$t = \frac{|b|}{|\operatorname{Im} a| + \sqrt{(\operatorname{Im} a)^2 - |b|^2}} \le \frac{8y}{7s} \le \frac{1}{4}.$$

Its hyperbolic distance from 0 is $\log \frac{1+t}{1-t} \le \frac{2t}{1-t} \le \frac{8}{3} t \le 12y/s$. □

Lemma 9.4 (Pair inequality). *Let G be a real symmetric positive semidefinite 2×2 matrix with trace S, and let R_ϱ be a rotation. Then $\Phi^*(G + R_\varrho^{\mathsf T} G R_\varrho)\Phi \ge S(1 - |\cos \varrho|)\|\Phi\|^2$ for every $\Phi \in \mathbb{C}^2$.*

Proof. Write $G = \frac{S}{2} I + g F_\phi$ with $F_\phi = \begin{pmatrix} \cos\phi & \sin\phi \\ \sin\phi & -\cos\phi \end{pmatrix}$ and $0 \le g \le S/2$. Since $R_\varrho^{\mathsf T} F_\phi R_\varrho = F_{\phi - 2\varrho}$ and $F_\phi + F_{\phi - 2\varrho} = 2\cos\varrho\, F_{\phi - \varrho}$, the real symmetric matrix $G + R_\varrho^{\mathsf T} G R_\varrho = SI + 2g\cos\varrho\, F_{\phi-\varrho}$ has eigenvalues $S \pm 2g|\cos\varrho| \ge S(1 - |\cos\varrho|)$. A real symmetric matrix bounds its Hermitian form on $\mathbb{C}^2$ below by its smallest eigenvalue. □

The point of Lemma 9.4 is that it holds for every polarization Φ. A single block may carry little mass, for instance when the solution is a standing wave with a node there, but two adjacent blocks cannot both do so.

Lemma 9.5 (An elliptic run). *Let $0 < \eta \le 1$ and assume $|\operatorname{tr} A_q(E, x_0)| \le 2 - \eta$. Let $B_0 \in SL(2,\mathbb{R})$ map i to the fixed point of $A_q(E, x_0)$ in the upper half-plane, and put*

$$S_0 = \sum_{r=0}^{q-1} \|e_1^{\mathsf T} A_r(E, x_0) B_0\|^2, \qquad \Phi_0 = B_0^{-1} \Psi_0,$$

$$\xi = (\mathrm{Lip}(v)+1)\max(\varepsilon,|\delta|), \qquad m_* = \Big\lfloor \frac{\eta^{3/2}}{2^{10}\xi S_0} \Big\rfloor .$$

If $m_* \ge 8$, then:

1. *for $0 \le m < m_*$, $\mathcal{M}_m + \mathcal{M}_{m+1} \ge 2^{-8}\eta\, S_0 \|\Phi_0\|^2$ and $\mathcal{M}_m \le 16 S_0 \|\Phi_0\|^2$;*
2. *with $m_0 = \lceil m_*/2 \rceil$,*
$$\sum_{k > m_0 q} |g(k)|^2 \ge \frac{\eta^{5/2}}{2^{22}} \cdot \frac{\|\Phi_0\|^2}{\xi};$$
3. *$\sum_{k>m_* q} |g(k)|^2 \le 2\|\Phi_0\|^2/\varepsilon$, $\sum_{k\ge 1} |g(k)|^2 \le (1/\xi + 2/\varepsilon)\|\Phi_0\|^2$, and $|g(0)| \le \|B_0\|\,\|\Phi_0\|$.*

By Lemma 8.2(1), $S_0 = |\partial_E \operatorname{tr} A_q(E,x_0)|/|\sin \varrho_0|$, where ϱ_0 is the rotation angle of $A_q(E,x_0)$. The run consists of m_* blocks, about $q/(\xi S_0)$ sites. Its length is set by the larger of the damping ε and the drift $|\delta|$; it is not a fixed function of the scale. When $\varepsilon \ge |\delta|$, the total mass in part 3 is at most $3(\mathrm{Lip}(v)+1)\|\Phi_0\|^2/\xi$, so part 2 says that the far half of the run carries a fixed fraction of the mass of g on $\mathbb{Z}_{>0}$. No arithmetic condition on θ enters, and nothing is assumed about the solution beyond the run.

Proof. Frames. Write $A_m = A_q(E,x_m)$ for the real-energy block monodromies. Suppose A_m is elliptic and $B_m \in SL(2,\mathbb{R})$ maps i to its fixed point, so that $B_m^{-1} A_m B_m = R_{\varrho_m}$ by Lemma 8.2(4). Put

$$u_r^{(m)} = e_1^{\mathsf T} A_r(E,x_m) B_m, \qquad G_m = \sum_{r=0}^{q-1} (u_r^{(m)})^{\mathsf T} u_r^{(m)}, \qquad S_m = \operatorname{tr} G_m, \qquad \Phi_m = B_m^{-1}\Psi_m .$$

One step. Apply Lemma 9.2 with $B = B_m$ and the shifts $w_r = v(x_m + r\alpha) - v(x_{m+1} + r\alpha)$, which satisfy $|w_r| \le \mathrm{Lip}(v)|\delta|$. It gives, for $0 \le r \le q$,

$$A_r(E,x_{m+1})B_m = A_r(E,x_m)B_m\,\Pi_r^{(m)}, \qquad \|\Pi_r^{(m)} - I\| \le y_m := e^{\mathrm{Lip}(v)|\delta| S_m} - 1.$$

In particular $B_m^{-1}A_{m+1}B_m = R_{\varrho_m}\Pi_q^{(m)}$ and $|\operatorname{tr} A_{m+1} - \operatorname{tr} A_m| \le 2y_m$. Put $s_0 = \sqrt{\eta}/2$. If $|\sin\varrho_m| \ge s_0$ and $y_m \le s_0/8$, Lemma 9.3 shows that A_{m+1} is elliptic with fixed point $B_m(w_m)$, where $d(i,w_m) \le d_m := 12 y_m/s_0$. Choose $N_m \in SL(2,\mathbb{R})$ with $N_m(i) = w_m$; by Lemma 8.2(4), $\|N_m\|^2 = \|N_m^{-1}\|^2 = e^{d(i,w_m)} \le e^{d_m}$. Then $B_{m+1} = B_m N_m$ maps i to the fixed point of A_{m+1}.

The damped step. Lemma 9.2 with $B = B_m$ and the constant shift $w_r = i\varepsilon$ gives $A_q(z,x_m)B_m = B_m R_{\varrho_m}\Pi'_m$ with $\det \Pi'_m = 1$ and $\|\Pi'_m - I\| \le e^{\varepsilon S_m} - 1$. Hence

$$\Phi_{m+1} = N_m^{-1} R_{\varrho_m} \Pi'_m\, \Phi_m .$$

Bootstrap. We show by induction that for $0 \le m \le m_*$ the frame B_m is defined and

$$(a_m) \qquad |\operatorname{tr} A_m| \le 2 - \eta/2, \qquad S_0/2 \le S_m \le 2S_0, \qquad \|\Phi_0\|/2 \le \|\Phi_m\| \le 2\|\Phi_0\|.$$

(a_0) holds. Assume (a_j) for all $j < m$, where $m \le m_*$. Note that $\varepsilon \le \xi$ and $\mathrm{Lip}(v)|\delta| \le \xi$, and that $m_* \ge 1$ forces $\xi S_0 \le 2^{-10}\eta^{3/2}$. For $j < m$ we have $|\cos\varrho_j| \le 1 - \eta/4$, hence $\sin^2\varrho_j \ge \eta/4$ and $|\sin\varrho_j| \ge s_0$. Also $\mathrm{Lip}(v)|\delta|S_j \le 2\xi S_0 \le 1$, and $e^x - 1 \le 2x$ for $0 \le x \le 1$, so

$$y_j \le 4\xi S_0 \le 2^{-8}\eta^{3/2} \le s_0/8.$$

So the frames $B_1, \dots, B_m$ are defined. Summing the one-step estimates over $j < m$:

- $|\operatorname{tr} A_m - \operatorname{tr} A_0| \le 2\sum_{j<m} y_j \le 8m_*\xi S_0 \le 2^{-7}\eta^{3/2} \le \eta/2$, so $|\operatorname{tr} A_m| \le 2 - \eta/2$.
- $u_r^{(j+1)} = u_r^{(j)}\Pi_r^{(j)}N_j$, so $S_{j+1}/S_j \in [(1-y_j)^2e^{-d_j}, (1+y_j)^2e^{d_j}]$ and $|\log(S_{j+1}/S_j)| \le 3y_j + d_j \le 27y_j/\sqrt{\eta}$. Hence $|\log(S_m/S_0)| \le 108\, m_*\xi S_0/\sqrt{\eta} \le 108\eta/2^{10} < \log 2$.
- A 2×2 matrix of determinant one has $\|M^{-1}\| = \|M\|$, and $\|\Pi_j'\| \le e^{\varepsilon S_j}$. So $|\log(\|\Phi_{j+1}\|/\|\Phi_j\|)| \le \varepsilon S_j + d_j/2 \le 2\xi S_0 + 12y_j/\sqrt{\eta} \le 50\xi S_0/\sqrt{\eta}$, and $|\log(\|\Phi_m\|/\|\Phi_0\|)| \le 50\eta/2^{10} < \log 2$.

This proves (a_m).

Pairs of blocks. Fix $0 \le m < m_*$ and put $\varepsilon_2 = e^{3\xi S_m} - 1$. Since $3\xi S_m \le 6\xi S_0 \le 1$, we have $\varepsilon_2 \le 12\xi S_0 \le 12\cdot 2^{-10}\eta^{3/2}$. By Lemma 9.2 with the shift $i\varepsilon$,

$$g(mq+1+r) = e_1^{\mathsf T}A_r(z,x_m)B_m\Phi_m = u_r^{(m)}(\Phi_m + e_r), \qquad \|e_r\| \le \varepsilon_2\|\Phi_m\|.$$

For the next block, $\Psi_{m+1} = B_mR_{\varrho_m}\Pi_m'\Phi_m$. Apply Lemma 9.2 first with the phase shifts, which gives $A_r(E,x_{m+1})B_m = A_r(E,x_m)B_m\Pi_r^{(m)}$, and then with the shift $i\varepsilon$, for which the rows $u_r^{(m)}\Pi_r^{(m)}$ have squared norms at most $(1+y_m)^2\|u_r^{(m)}\|^2 \le 2\|u_r^{(m)}\|^2$. This gives

$$g((m+1)q+1+r) = u_r^{(m)}\,\Pi_r^{(m)}\Pi_r''\,R_{\varrho_m}\Pi_m'\,\Phi_m = u_r^{(m)}(R_{\varrho_m}\Phi_m + e_r'),$$

where $\|\Pi_r'' - I\| \le e^{2\varepsilon S_m} - 1$. The three factors $\Pi_r^{(m)}$, Π_r'' and Π_m' differ from I by at most $e^{\operatorname{Lip}(v)|\delta|S_m} - 1$, $e^{2\varepsilon S_m} - 1$ and $e^{\varepsilon S_m} - 1$, and R_{ϱ_m} is orthogonal. Hence $\|\Pi_r^{(m)}\Pi_r''R_{\varrho_m}\Pi_m' - R_{\varrho_m}\| \le e^{(\operatorname{Lip}(v)|\delta| + 3\varepsilon)S_m} - 1 \le \varepsilon_2$, because $\operatorname{Lip}(v)|\delta| + 3\varepsilon \le 3\xi$. So $\|e_r'\| \le \varepsilon_2\|\Phi_m\|$.

For every $e \in \mathbb{C}^2$, $\sum_r |u_r^{(m)}e|^2 = e^*G_me \le S_m\|e\|^2$, and $\Phi^*G_m\Phi = \sum_r |u_r^{(m)}\Phi|^2$. Apply $\|a+b\|^2 \ge \frac12\|a\|^2 - \|b\|^2$ to vectors indexed by $0 \le r < q$:

$$\mathcal{M}_m \ge \tfrac12\Phi_m^*G_m\Phi_m - S_m\varepsilon_2^2\|\Phi_m\|^2, \qquad \mathcal{M}_{m+1} \ge \tfrac12\Phi_m^*R_{\varrho_m}^{\mathsf T}G_mR_{\varrho_m}\Phi_m - S_m\varepsilon_2^2\|\Phi_m\|^2.$$

By (a_m), $|\cos\varrho_m| \le 1 - \eta/4$, so Lemma 9.4 and $2\varepsilon_2^2 \le \eta/16$ give

$$\mathcal{M}_m + \mathcal{M}_{m+1} \ge \Big(\frac{\eta}{8} - 2\varepsilon_2^2\Big)S_m\|\Phi_m\|^2 \ge \frac{\eta}{16}S_m\|\Phi_m\|^2 \ge \frac{\eta}{2^7}S_0\|\Phi_0\|^2.$$

Also $\mathcal{M}_m \le (1+\varepsilon_2)^2S_m\|\Phi_m\|^2 \le 16S_0\|\Phi_0\|^2$. This proves part 1.

Flux. Multiply (16) by $\overline{g(k)}$, sum over $k > n$ for some $n \ge 0$, and take imaginary parts. Since $g \in \ell^2$ and $\sum_{k>n} 2\operatorname{Re}\big(g(k+1)\overline{g(k)}\big)$ is real,

$$\varepsilon\sum_{k>n}|g(k)|^2 = \operatorname{Im}\big(g(n)\overline{g(n+1)}\big).$$

If $(g(n+1), g(n))^{\mathsf T} = B\Phi$ with B real and $\det B = 1$, then $2|\operatorname{Im}(g(n)\overline{g(n+1)})| = |\det(B\Phi, B\bar\Phi)| = |\det(\Phi,\bar\Phi)| \le \|\Phi\|^2$. At $n = m_*q$ this gives $\sum_{k>m_*q}|g(k)|^2 \le \|\Phi_{m_*}\|^2/(2\varepsilon) \le 2\|\Phi_0\|^2/\varepsilon$.

Counting. Let $X = \eta^{3/2}/(2^{10}\xi S_0)$, so that $X \ge m_* \ge 8$ and $m_* \ge X/2$. The blocks $m_0, \dots, m_*$ lie in $\{k > m_0q\}$ and contain at least $\lfloor m_*/2\rfloor/2 \ge m_*/8$ disjoint pairs $(m, m+1)$ with $m < m_*$. By part 1,

$$\sum_{k>m_0q}|g(k)|^2 \ge \frac{m_*}{8}\cdot\frac{\eta}{2^8}S_0\|\Phi_0\|^2 \ge \frac{X}{16}\cdot\frac{\eta}{2^8}S_0\|\Phi_0\|^2 = \frac{\eta^{5/2}}{2^{22}}\cdot\frac{\|\Phi_0\|^2}{\xi}.$$

For part 3, $\sum_{k\ge1}|g(k)|^2 = \sum_{m<m_*}\mathcal{M}_m + \sum_{k>m_*q}|g(k)|^2$, and $16m_*S_0 \le 16XS_0 = 2^{-6}\eta^{3/2}/\xi \le 1/\xi$. Finally $g(0) = e_2^{\mathsf T}\Psi_0 = e_2^{\mathsf T}B_0\Phi_0$. □

9.3 A fixed fraction at resonant times

From now on $v(x) = 2\lambda\cos 2\pi x$ again, so $\mathrm{Lip}(v) = 4\pi\lambda$. We first compare the traces of the actual q-blocks with the rational discriminant.

Lemma 9.6 (Block traces)**.** *For every $h > L$ there is C such that for all large n, all real $E \in U_h$ and all $x \in \mathbb{T}$,*

$$\left|\operatorname{tr} A_{q_n}(E,x) - P_{q_n}(E) + 2\lambda^{q_n}\cos(2\pi q_n x)\right| \le C\, q_n e^{hq_n}|\delta_n|.$$

Proof. By Lemma 3.1, $P_q(E) - 2\lambda^q\cos(2\pi qx) = \operatorname{tr} A_q^{(p/q)}(E,x)$. As in Step 1 of the proof of Proposition 3.3, write each rational factor as the irrational one plus an error of norm at most $d = 4\pi\lambda|\delta|$ and expand. The difference $A_q^{(p/q)}(E,x) - A_q(E,x)$ is the sum of the terms with at least one error, so its norm is at most

$$\sum_{k=1}^{q}\binom{q}{k}C_h^{k+1}e^{h(q-k)}d^k = C_h e^{hq}\left[(1+C_h d e^{-h})^q - 1\right] \le C_h^2 qd\, e^{hq}e^{C_h qd},$$

and $qd \le 4\pi\lambda$. The trace of a 2×2 matrix is at most twice its norm. □

Theorem 9.7 (A fixed fraction moves out)**.** *Let $\lambda > 1$, let α be irrational and let θ be any phase. Fix $h > L$, $c > h$ and $0 < \eta \le 1/4$, and put*

$$f_0 = \frac{\eta^{5/2}}{2^{25}(4\pi\lambda+1)}.$$

There is n_0, depending only on λ, α, h, c and η, such that for every $n \ge n_0$ with $b_n \ge c$ and every T with $e^{hq_n} \le T \le |\delta_n|^{-1}$,

$$P\big(Te^{-hq_n}, T\big) \ge f_0\,\mu^{\theta}\big(\{E : |\Delta_n(E)| \le 2 - 3\eta\}\big).$$

If $b_n \ge c$ for infinitely many n, then along those n the right side tends to $f_0\frac{2}{\pi}\arcsin(1-\frac{3\eta}{2})$, by Lemma 9.1. At the resonant time $T = |\delta_n|^{-1} = e^{b_n q_n}$ the distance is $e^{(b_n-h)q_n} = T^{1-h/b_n}$.

Proof. Let $h_0 = (L+h)/2$ and $U = U_{h_0}$. By Lemma 2.3 and Proposition 3.3(2) there is C with $\|A_r(E,x)\| \le Ce^{h_0 r}$ for real $E \in U$, all x and $r \ge 0$, and $\sup_U |P_q'| \le Cqe^{h_0 q}$. Write $q = q_n$, $\delta = \delta_n$, $b = b_n$ and $\varepsilon = 1/T$, so that $|\delta| \le \varepsilon \le e^{-hq}$, and put

$$N = Te^{-hq}, \qquad I = \{E \in U :\ |\Delta_n(E)| \le 2 - 2\eta\}.$$

Below, C and c' denote positive constants depending only on λ, α, h and η, and "for large n" means for n beyond a threshold depending only on λ, α, h, c and η; nothing depends on θ or T.

Step 1: two elliptic first blocks. Fix $E \in I$ and $z = E + i\varepsilon$. For $k \ge 0$ put $g_+(k) = G(k;z)$ and $g_-(k) = G(-k;z)$. Both lie in $\ell^2(\mathbb{Z}_{\ge 0})$. The function g_+ satisfies (16) for the phase θ. Since the cosine is even, $v(\theta - k\alpha) = v(-\theta + k\alpha)$, so g_- satisfies (16) for the phase $-\theta$. The first blocks are $A_q(E, \pm\theta+\alpha)$. Since $q(\pm\theta+\alpha) = \pm q\theta + p + \delta$, Lemma 9.6 and $|\cos(2\pi(\pm q\theta+\delta)) - \cos(2\pi q\theta)| \le 2\pi|\delta|$ give

$$\left|\operatorname{tr} A_q(E, \pm\theta+\alpha) - \Delta_n(E)\right| \le Cqe^{h_0 q}|\delta| + 4\pi\lambda^q|\delta|.$$

The right side tends to 0 because $b \ge c > h_0 > L$. So for large n both first blocks satisfy $|\operatorname{tr} A_q| \le 2 - \eta$.

Step 2: two elliptic runs. Apply Lemma 9.5 to $g_\pm$, with this η, the damping ε and $\xi = (4\pi\lambda+1)\varepsilon$, and let $B_0^\pm$, $S_0^\pm$, $\Phi_0^\pm$, $m_*^\pm$, $m_0^\pm$ be the corresponding data. From $|\operatorname{tr} A_q| \le 2-\eta$ we get $|\sin \varrho_0^\pm| \ge \sqrt{\eta}/2$, and the product rule gives $|\partial_E \operatorname{tr} A_q(E,x)| \le 2qC^2 e^{h_0 q}$. By Lemma 8.2(1), $S_0^\pm \le Cqe^{h_0 q}$. Hence

$$\frac{\eta^{3/2}}{2^{10}\xi S_0^\pm} \ge \frac{c'T}{qe^{h_0 q}} \ge \frac{c'e^{(h-h_0)q}}{q},$$

so for large n we have $m_*^\pm \ge 8$, and $m_0^\pm q \ge m_*^\pm q/2 \ge c'Te^{-h_0 q}/4 \ge Te^{-hq} = N$. Lemma 9.5 and $\varepsilon \ge |\delta|$ give

$$\sum_{k>N} |g_\pm(k)|^2 \ge \frac{\eta^{5/2}}{2^{22}} \cdot \frac{\|\Phi_0^\pm\|^2}{\xi}, \qquad \sum_{k\ge 1} |g_\pm(k)|^2 \le \frac{3(4\pi\lambda+1)\|\Phi_0^\pm\|^2}{\xi}.$$

The site 0 is negligible. By Lemma 8.2(4), $\|B_0^+\|^2 = e^{d(i,z_+)}$, where z_+ is the fixed point of $\mathcal{A} = A_q(E,\theta+\alpha)$. The identity $\|\mathcal{A}\|_{\mathrm{HS}}^2 = 2+4\sinh^2(d(i,z_+))\sin^2\varrho_0^+$ from the proof of Lemma 8.2(4) gives $\|B_0^+\|^2 \le 1 + 4\|\mathcal{A}\|/\sqrt{\eta} \le Ce^{h_0 q}$. With part 3 of Lemma 9.5,

$$|G(0;z)|^2 \le \|B_0^+\|^2\|\Phi_0^+\|^2 \le Ce^{h_0 q}\xi \sum_{k>N} |g_+(k)|^2,$$

and $e^{h_0 q}\xi \le (4\pi\lambda+1)e^{(h_0-h)q} \to 0$. Let $\mathcal{F} = \sum_{|k|>N} |G(k;z)|^2$, the left side of (17) below. The two displays give $\sum_{k\ne 0} |G(k;z)|^2 \le 3\cdot 2^{22}(4\pi\lambda+1)\eta^{-5/2}\mathcal{F}$, and $|G(0;z)|^2 \le o(1)\mathcal{F}$. Hence, for large n,

$$\sum_{|k|>N} |G(k;z)|^2 \ge \frac{\eta^{5/2}}{2^{24}(4\pi\lambda+1)} \sum_{k\in\mathbb{Z}} |G(k;z)|^2 = 2f_0 \sum_{k\in\mathbb{Z}} |G(k;z)|^2 \qquad (E\in I). \tag{17}$$

Step 3: Parseval. By (14), (17) and (15),

$$P(N,T) = \frac{1}{\pi T}\int_{\mathbb{R}} \sum_{|k|>N} |G(k;E+i\varepsilon)|^2\, dE \ge 2f_0 \int_I \frac{1}{\pi}\operatorname{Im} G(0;E+i\varepsilon)\, dE,$$

and the right side equals $2f_0 \int_{\mathbb{R}} \bigl(\int_I P_\varepsilon(E-E')\,dE\bigr)\, d\mu^\theta(E')$.

Step 4: Poisson smoothing. Let $r_* > 0$ be the distance from Σ to $\mathbb{R}\setminus U$, and $r = \min(r_*/2, \eta/\sup_U |P_q'|)$, so that $r \ge \eta/(Cqe^{h_0 q})$ for large n. If $E'\in\Sigma$ and $|\Delta_n(E')| \le 2-3\eta$, then $[E'-r, E'+r] \subset U$, and $\Delta_n' = P_q'$ gives $|\Delta_n| \le 2-2\eta$ there. So $[E'-r,E'+r]\subset I$ and

$$\int_I P_\varepsilon(E-E')\,dE \ge \frac{2}{\pi}\arctan\frac{r}{\varepsilon} \ge 1 - \frac{2\varepsilon}{\pi r}.$$

Since $\varepsilon/r \le Cqe^{h_0 q}/(\eta T) \le Cqe^{(h_0-h)q}/\eta \to 0$ and μ^θ is supported on Σ, we obtain $P(N,T) \ge 2f_0(1-o(1))\,\mu^\theta(\{|\Delta_n| \le 2-3\eta\})$, which is at least $f_0\,\mu^\theta(\{|\Delta_n| \le 2-3\eta\})$ for large n. □

Remark 9.8 (An annulus at strongly resonant scales)**.** Theorem 9.7 has a matching upper half. Let $\theta \in G_\alpha$ and $\beta(\alpha) < \infty$, and fix $B_0 > \beta(\alpha)$, $h > L$, $c > h$, $\varepsilon > 0$ and $M > 0$. For large n with $b_n \ge c$ we have $b_n \le B_0$. For every T with $e^{hq_n} \le T \le |\delta_n|^{-1}$ put

$$N_- = Te^{-hq_n} = T\lambda^{-q_n}e^{-(h-L)q_n}, \qquad N_+ = e^{\varepsilon q_n}\bigl(1+T\lambda^{-q_n}\bigr) \le 2e^{\varepsilon q_n}T\lambda^{-q_n}.$$

Theorem 5.4 with $\Lambda = \max(1, B_0)$ gives $P(N_+,T) \le Ce^{-Mq_n}$, and Theorem B, which is deduced from Theorem 9.7 in Section 9.4, gives $P(N_-,T) \ge c_\lambda := 2^{-32}/(1+4\pi\lambda)$. So the annulus $N_- < |j| \le N_+$

carries Abel-averaged mass at least $c_\lambda - Ce^{-Mq_n}$. In words: a fixed positive fraction of the Abel-averaged packet lies at distance $T\lambda^{-q_n}$, up to the factors $e^{(h-L)q_n}$ and $2e^{\varepsilon q_n}$, and the mass beyond N_+ is exponentially small. The exponent tolerances $h - L$ and ε can be fixed as small as we like before n is taken large; this is the sense in which the factors are $e^{o(q_n)}$. The estimates say nothing about how the remaining mass, up to $1 - c_\lambda$, is distributed inside N_-, and they do not locate a median of the packet. The window begins at $T = e^{hq_n}$, slightly after λ^{q_n}. All of this concerns time averages, not the instantaneous position of the packet.

9.4 Every moment

Corollary 9.9. *Let $\lambda > 1$ and let α be irrational. For every phase θ and every $p > 0$,*

$$\beta^+(p) \geq \max\Big\{0,\ 1 - \frac{\log\lambda}{\beta(\alpha)}\Big\},$$

where the right side is read as 0 when $\beta(\alpha) = 0$ and as 1 when $\beta(\alpha) = \infty$. If $\beta(\alpha) > \log\lambda$, moreover $\limsup_{T\to\infty} P(T^a, T) > 0$ for every $0 < a < 1 - \log\lambda/\beta(\alpha)$.

Proof. Nonnegativity. The vector δ_0 is not an eigenvector of H, since $H\delta_0$ has nonzero entries at ± 1. So μ^θ is not a point mass and $\sum_E \mu^\theta(\{E\})^2 < 1$. Now

$$a(0,T) = \iint \frac{2/T}{2/T + i(E - E')}\, d\mu^\theta(E)\, d\mu^\theta(E'),$$

and the integrand has modulus at most one and tends to $\mathbf{1}_{E=E'}$, so $a(0,T) \to \sum_E \mu^\theta(\{E\})^2$ by dominated convergence. Hence $M_p(T) \geq P(0,T) = 1 - a(0,T)$ is bounded below for large T, and $\beta^+(p) \geq 0$.

The main bound. Assume $\beta(\alpha) > L$. Fix $h \in (L, \beta(\alpha))$ and $c \in (h, \beta(\alpha))$. Since $\limsup b_n = \beta(\alpha)$, there are $n_j \to \infty$ with $b_{n_j} \geq c$ and $b_{n_j} \to \beta(\alpha)$. Apply Theorem 9.7 with $\eta = 1/4$ at $T_j = |\delta_{n_j}|^{-1}$, and put $N_j = T_j e^{-hq_{n_j}} = e^{(b_{n_j} - h)q_{n_j}}$. By Lemma 9.1, $\mu^\theta(\{|\Delta_{n_j}| \leq 5/4\}) \to \frac{2}{\pi}\arcsin\frac{5}{8} > \frac{1}{4}$, so $P(N_j, T_j) \geq c_0 := f_0/4$ for large j. Hence

$$M_p(T_j) \geq N_j^p\, P(N_j, T_j) \geq c_0 N_j^p, \qquad \frac{\log N_j}{\log T_j} = \frac{b_{n_j} - h}{b_{n_j}} \to 1 - \frac{h}{\beta(\alpha)},$$

with limit 1 if $\beta(\alpha) = \infty$. So $\beta^+(p) \geq 1 - h/\beta(\alpha)$, and letting $h \downarrow L$ proves the bound. If $a < 1 - L/\beta(\alpha)$, choose h with $a < 1 - h/\beta(\alpha)$; then $T_j^a \leq N_j$ for large j, so $P(T_j^a, T_j) \geq c_0$. □

Damanik and Tcheremchantsev [11] also use the exponent α_l^+, the supremum of the a for which $P(T^a, T)$ decays at no polynomial rate along some sequence of times; always $\alpha_l^+ \leq \alpha_u^+$. The second statement of Corollary 9.9 gives $\alpha_l^+ \geq 1 - \log\lambda/\beta(\alpha)$. So for $\theta \in G_\alpha$ and $\beta(\alpha) < \infty$, $\alpha_l^+ = \alpha_u^+$: the fastest tail is carried by a fixed fraction of the packet.

Proof of Theorem A. Part 1. Theorem 7.3 gives $P(T^a, T) = O(T^{-m})$ for every $a > \max\{0, 1 - L/\beta(\alpha)\}$ and every $m > 0$, and $\alpha_u^+ \leq \max\{0, 1 - L/\beta(\alpha)\}$. Since $\beta^+(p) \leq \alpha_u^+$ by (4), the same bound holds for every $\beta^+(p)$. Corollary 9.9 gives the reverse inequality $\beta^+(p) \geq \max\{0, 1 - L/\beta(\alpha)\}$ at every phase, and $\alpha_u^+ \geq \beta^+(p)$.

Part 2. Corollary 9.9 gives $\beta^+(p) \geq 1$ at every phase, and Lemma 2.2(1) gives $\beta^+(p) \leq 1$. Hence $\beta^+(p) = \alpha_u^+ = 1$. This recovers [18, Corollary 3]. □

Proof of Theorem B. Take $\eta = 1/4$ in Theorem 9.7, so $f_0 = 2^{-30}/(4\pi\lambda + 1)$. By Lemma 9.1, along the indices with $b_n \geq c$ we have $\mu^\theta(\{|\Delta_n| \leq 5/4\}) \to \frac{2}{\pi}\arcsin\frac{5}{8} \geq \frac{5}{4\pi} > \frac{1}{4}$. So for all large such n and all $T \in [e^{hq_n}, |\delta_n|^{-1}]$, $P(Te^{-hq_n}, T) \geq f_0/4 = 2^{-32}/(1 + 4\pi\lambda)$. □

9.5 Local dimensions

A lower bound on transport does not by itself bound packing dimensions from below: the inequalities of Guarneri and Schulz-Baldes go the other way. The lower bound on dimensions comes instead from a pointwise estimate on the resolvent, which the elliptic runs also provide.

Lemma 9.10 (Local mass)**.** *Let θ, h, c and η be as in Theorem 9.7, and put $h_0 = (L+h)/2$. There are C and n_0, depending only on λ, α, h, c and η, such that for every $n \ge n_0$ with $b_n \ge c$ and every $E \in U_{h_0}$ with $|\Delta_n(E)| \le 2 - 2\eta$,*

$$|G(0; E + i|\delta_n|)| \le Ce^{h_0 q_n}, \qquad \mu^\theta([E - |\delta_n|, E + |\delta_n|]) \le 2Ce^{h_0 q_n}|\delta_n|.$$

Proof. Use Steps 1 and 2 of the proof of Theorem 9.7 with $T = |\delta_n|^{-1}$, so $\varepsilon = |\delta_n|$ and $\xi = (4\pi\lambda+1)\varepsilon$, on the right half-line only. By (15) and Lemma 9.5(2),

$$\operatorname{Im} G(0; z) = \varepsilon \sum_k |G(k; z)|^2 \ge \varepsilon \cdot \frac{\eta^{5/2}}{2^{22}} \cdot \frac{\|\Phi_0^+\|^2}{\xi} = c_1 \|\Phi_0^+\|^2, \qquad c_1 = \frac{\eta^{5/2}}{2^{22}(4\pi\lambda + 1)}.$$

By Lemma 9.5(3), $\operatorname{Im} G(0; z) \le |G(0; z)| \le \|B_0^+\| \, \|\Phi_0^+\|$. Hence $\|\Phi_0^+\| \le \|B_0^+\|/c_1$ and $|G(0; z)| \le \|B_0^+\|^2/c_1 \le Ce^{h_0 q}$, by the bound on $\|B_0^+\|^2$ in Step 2. Finally $\operatorname{Im} G(0; E + i\varepsilon) \ge \int_{|E'-E|\le\varepsilon} \frac{\varepsilon}{2\varepsilon^2} \, d\mu^\theta(E')$, so $\mu^\theta([E - \varepsilon, E + \varepsilon]) \le 2\varepsilon \, |G(0; E + i\varepsilon)|$. □

Proof of Theorem C, part 1. Let $\beta = \beta(\alpha)$. Upper local dimensions of a finite measure are nonnegative, so assume $\beta > L$. Fix $h \in (L, \beta)$, $c \in (h, \beta)$ and $n_j \to \infty$ with $b_{n_j} \ge c$ and $b_{n_j} \to \beta$, and put $h_0 = (L + h)/2$, $U = U_{h_0}$ and $r_j = |\delta_{n_j}|$.

Two canonical measures at the same radii. Let $\theta' = \theta + \alpha$. Translation by one site conjugates $H_{\lambda,\alpha,\theta}$ to $H_{\lambda,\alpha,\theta'}$ and maps δ_1 to δ_0, as in the proof of Proposition 9.11 below. So $\mu_{\delta_1} = \mu^{\theta'}$, and

$$\nu := \mu_{\delta_0} + \mu_{\delta_1} = \mu^\theta + \mu^{\theta'}.$$

Write Δ_n and Δ_n' for the discriminants at the phases θ and θ'. Since $q_n\theta' = q_n\theta + p_n + \delta_n$, we have $|\Delta_n' - \Delta_n| \le 4\pi\lambda^{q_n}|\delta_n| \to 0$. For $0 < \eta \le 1/8$ let $J_j(\eta) = \{E \in U : |\Delta_{n_j}(E)| \le 2 - 4\eta\}$. For large j, every $E \in J_j(\eta)$ satisfies $|\Delta_{n_j}(E)| \le 2 - 2\eta$ and $|\Delta_{n_j}'(E)| \le 2 - 2\eta$. Lemma 9.10, applied at the phases θ and θ', then gives

$$\nu([E - r_j, E + r_j]) \le 4Ce^{h_0 q_{n_j}} r_j \qquad (E \in J_j(\eta), \ j \text{ large}).$$

A full-measure set. Let $\Lambda_\eta = \limsup_j J_j(\eta)$. Each $E \in \Lambda_\eta$ lies in $J_j(\eta)$ for infinitely many j, and $\log r_j = -b_{n_j} q_{n_j}$, so

$$\frac{\log \nu([E - r_j, E + r_j])}{\log r_j} \ge 1 - \frac{h_0}{b_{n_j}} - \frac{\log(4C)}{b_{n_j} q_{n_j}} \to 1 - \frac{h_0}{\beta}$$

along those j, with limit 1 if $\beta = \infty$. Thus $d_\nu^+ \ge 1 - h/\beta$ on Λ_η. For a finite measure, $\nu(\limsup_j J_j) \ge \limsup_j \nu(J_j)$. The measures μ^θ and $\mu^{\theta'}$ are supported on $\Sigma \subset U$, and for large j the set $J_j(\eta)$ contains $\{E \in \Sigma : |\Delta_{n_j}'(E)| \le 2 - 5\eta\}$. Lemma 9.1 at the two phases gives

$$\nu(\Lambda_\eta) \ge \frac{2}{\pi} \arcsin(1 - 2\eta) + \frac{2}{\pi} \arcsin\Big(1 - \frac{5\eta}{2}\Big),$$

which tends to $2 = \nu(\mathbb{R})$ as $\eta \to 0$. So $\bigcup_{k \ge 8} \Lambda_{1/k}$ has full ν-measure, and $d_\nu^+ \ge 1 - h/\beta$ holds ν-almost everywhere. Letting $h \downarrow L$ along a sequence gives $d_\nu^+ \ge \max\{0, 1 - L/\beta\}$ ν-almost everywhere.

Every spectral measure. Since $\delta_{k+1} = (H - v(\theta + k\alpha))\delta_k - \delta_{k-1}$, the vectors δ_0, δ_1 generate $\ell^2(\mathbb{Z})$ under H, and every spectral measure μ_φ is absolutely continuous with respect to ν. Let $f = d\mu_\varphi/d\nu$. By the differentiation theorem for Radon measures [20, Theorem 2.12], $\mu_\varphi([E-r, E+r])/\nu([E-r, E+r]) \to f(E)$ as $r \downarrow 0$ for ν-almost every E, and $0 < f(E) < \infty$ for μ_φ-almost every E. At such points $\log \mu_\varphi([E-r, E+r]) = \log \nu([E-r, E+r]) + O(1)$, so $d^+_{\mu_\varphi}(E) = d^+_\nu(E)$. Since $\mu_\varphi \ll \nu$, the bound $d^+_\nu \geq \max\{0, 1 - L/\beta\}$ holds μ_φ-almost everywhere. □

The upper bound in part 2 is a consequence of the upper transport bound, Theorem 7.3.

Proposition 9.11. *Let $\lambda > 1$ and let α be irrational with $\beta(\alpha) < \infty$. For every $\theta \in G_\alpha \cap (G_\alpha - \alpha)$ and every $\varphi \in \ell^2(\mathbb{Z})$, the spectral measure μ_φ of $H_{\lambda,\alpha,\theta}$ satisfies $\dim^+_P(\mu_\varphi) \leq \max\{0, 1 - \log\lambda/\beta(\alpha)\}$.*

Proof. Guarneri and Schulz-Baldes [13, Theorem 1] proved that for every $q > 0$ the upper growth exponent of the q-th moment, computed with the Cesàro average $T^{-1}\int_0^T \cdot\, dt$, is at least the upper packing dimension $\dim^+_P(\mu_{\delta_0})$, defined as the essential supremum of the upper local dimension. The same inequality holds for every unit vector $\psi \in \ell^2(\mathbb{Z})$ whose spectral measure has compact support, by [3, Theorem 1 and Proposition 1(iii)]: with $\langle\langle |X|^p \rangle\rangle_\psi(T) = T^{-1}\int_0^T \||X|^{p/2} e^{-itH}\psi\|^2\, dt$,

$$\limsup_{T\to\infty} \frac{\log\langle\langle |X|^p\rangle\rangle_\psi(T)}{\log T} \geq p\, D^+_{\mu_\psi}\Big(\frac{1}{1+p}\Big) \geq p \dim^+_P(\mu_\psi) \qquad (p > 0),$$

where D^+_μ is the upper generalized Rényi dimension. Since $(2/T)e^{-2t/T} \geq (2/T)e^{-2}$ for $0 \leq t \leq T$, the Abel-averaged moment $M_q(T)$ dominates $2e^{-2}$ times the Cesàro-averaged one. So the Cesàro exponents are at most $\beta^+(q) \leq \alpha^+_u$, and Theorem 7.3 gives $\dim^+_P(\mu_{\delta_0}) \leq \max\{0, 1 - L/\beta\}$ for $\theta \in G_\alpha$.

The translation $(Uu)(n) = u(n+1)$ satisfies $UH_{\lambda,\alpha,\theta}U^{-1} = H_{\lambda,\alpha,\theta+\alpha}$ and $U\delta_1 = \delta_0$. Hence the spectral measure of δ_1 for the phase θ equals that of δ_0 for the phase $\theta + \alpha$, and the same bound holds when $\theta + \alpha \in G_\alpha$. For a Jacobi operator every spectral measure μ_φ is absolutely continuous with respect to $\nu = \mu_{\delta_0} + \mu_{\delta_1}$. The upper packing dimension of a measure is the infimum of the packing dimensions of its full-measure Borel sets [13, Appendix]. Packing dimension of sets is countably stable, so $\dim^+_P(\nu) = \max\{\dim^+_P(\mu_{\delta_0}), \dim^+_P(\mu_{\delta_1})\}$, and absolute continuity gives $\dim^+_P(\mu_\varphi) \leq \dim^+_P(\nu)$. □

Proof of Theorem C, part 2. By Proposition 9.11, the μ_φ-essential supremum of $d^+_{\mu_\varphi}$ is at most $\max\{0, 1 - L/\beta(\alpha)\}$. By part 1, $d^+_{\mu_\varphi}$ is at least this number μ_φ-almost everywhere. So $d^+_{\mu_\varphi}$ equals it μ_φ-almost everywhere, and its essential infimum $\dim^-_P(\mu_\varphi)$ and essential supremum $\dim^+_P(\mu_\varphi)$ both equal it. The set $G_\alpha \cap (G_\alpha - \alpha)$ has full measure by Lemma 2.1. □

10 Computability

We use standard notions for reals: a real is computable from an oracle if the oracle computes a sequence of rationals converging to it with a known rate, and two reals have the same Turing degree if each is computable from the other. A real x is right-c.e. relative to $\mathbf{0}'$ if $\{r \in \mathbb{Q} : r > x\}$ is c.e. relative to $\mathbf{0}'$. We use one external lemma.

Lemma 10.1 (Becher, Bugeaud and Slaman [4, Lemma 4])**.** *A real x is the* lim sup *of a computable sequence of rationals if and only if x is right-c.e. relative to $\mathbf{0}'$.*

Proof of Theorem D. 1. By Theorem A, $\alpha_u^+ = \max\{0, 1 - L/\beta\}$, a computable function of β when λ is computable. If $\beta > L$ then $\beta = L/(1 - \alpha_u^+)$, so β is computable from α_u^+.

2. Let $\lambda = 2$, so $L = \log 2$, and $\theta = 1/4$. If the denominators of α are eventually odd, then $|\sin(2\pi q_n/4)| = 1$ eventually, so $1/4 \in G_\alpha$ and Theorem A applies.

Every value is right-c.e. relative to $\mathbf{0}'$. For computable α the ratios β_n form a computable sequence of reals, so rationals within 2^{-n} of them form a computable sequence with the same lim sup. If $\beta = \infty$ then $\alpha_u^+ = 1$. Otherwise β is right-c.e. relative to $\mathbf{0}'$ by Lemma 10.1, and so is $x = \max\{0, 1 - L/\beta\}$: for rational $r \in (0, 1)$, $x < r$ holds if and only if some rational s satisfies $\beta < s < L/(1 - r)$, which is a $\mathbf{0}'$-c.e. condition; $x < r$ never holds for $r \le 0$ and always holds for $r \ge 1$.

Every such value occurs. For $x = 1$ take $a_1 = 1$ and $a_{n+1} = 2^{q_n^2}$, so $\beta = \infty$. For $x = 0$ take $a_1 = 1$ and $a_n = 2$ for $n \ge 2$, so $\beta = 0$. Let $x \in (0, 1)$ be right-c.e. relative to $\mathbf{0}'$. Then $s = 1/(1 - x) > 1$ is right-c.e. relative to $\mathbf{0}'$, by the same monotonicity argument. Lemma 10.1 gives a computable sequence of rationals v_n with $\limsup v_n = s$; replacing v_n by $\min\{\max\{v_n, 1\}, B\}$ for an integer $B > s$ does not change the lim sup. Define

$$a_1 = 1, \qquad a_{n+1} = 2^{\lceil v_n q_n \rceil} \quad (n \ge 1).$$

Since $a_{n+1} q_n \le q_{n+1} \le 2 a_{n+1} q_n$,

$$0 \le \beta_n - (\log 2) v_n \le \frac{\log(4 q_n)}{q_n},$$

so $\beta = s \log 2 > L$ and $\alpha_u^+ = 1 - L/\beta = 1 - 1/s = x$. In all three constructions every partial quotient after a_1 is even, so $q_0 = q_1 = 1$ and the recurrence $q_{n+1} = a_{n+1} q_n + q_{n-1}$ keeps every q_n odd. The partial quotients are computable, and $|\alpha - p_n/q_n| < q_n^{-2}$ with $q_n \to \infty$, so α is computable.

3. Let K_2 be the halting problem relative to $\mathbf{0}'$, a set of degree $\mathbf{0}''$ that is c.e. relative to $\mathbf{0}'$, and put

$$s = 2 - \sum_{e \in K_2} 4^{-(e+2)} \in \Big[\frac{23}{12}, 2\Big].$$

The sum is left-c.e. relative to $\mathbf{0}'$, so s is right-c.e. relative to $\mathbf{0}'$, and part 2 produces a computable α with $\alpha_u^+(H_{2,\alpha,1/4}) = x = 1 - 1/s \in [11/23, 1/2]$. The oracle K_2 computes s. Conversely, from s one recovers K_2: the base-4 digits of $2 - s$ are 0 or 1, and after the digits at positions below $e + 2$ are subtracted, the remainder is at most $\frac{1}{3} \cdot 4^{-(e+2)}$ if $e \notin K_2$ and at least $4^{-(e+2)}$ if $e \in K_2$, so a sufficiently good rational approximation decides membership. Thus s, and hence $x = 1 - 1/s$, has degree exactly $\mathbf{0}''$. □

Remark 10.2. When the ratios β_n converge, their limit is computable from $\mathbf{0}'$, and every nonnegative real t computable from $\mathbf{0}'$ arises as such a limit for a computable α with odd denominators. By the limit lemma there are computable rationals $w_n \to t/\log 2$; put $a_1 = 1$ and $a_{n+1} = 2^{\max\{1, \lceil q_n \max\{w_n, 0\} \rceil\}}$, so that the partial quotients are even and $\beta_n \to t$, including $t = 0$. In this regular class the exponent therefore has degree at most $\mathbf{0}'$, and degree $\mathbf{0}'$ occurs. For instance, with $h_K = \sum_{j \in K} 2 \cdot 4^{-(j+1)}$ for the halting set K and v_n the stage-n approximations of $3/2 + h_K/4$, the exponent of $H_{2,\alpha,1/4}$ is $1 - 1/s \in [1/3, 2/5]$ with $s = 3/2 + h_K/4$, of degree $\mathbf{0}'$. Part 3 of Theorem D needs ratios that oscillate: a convergent computable sequence cannot have a limit of degree $\mathbf{0}''$.

Remark 10.3. All three parts of Theorem D hold for every moment exponent: by Theorem A, $\beta^+(p) = \alpha_u^+$ for every $p > 0$ in parts 1–3. They also hold for packing dimensions, with the following phase conditions.

In part 1, $\theta \in G_\alpha$. The first paragraph of the proof of Proposition 9.11 gives $\dim_P^+(\mu_{\delta_0}) \le \alpha_u^+$, and Theorem C(1) gives $d_{\mu_{\delta_0}}^+ \ge \alpha_u^+$ almost everywhere. So the spectral measure of δ_0 has $\dim_P^- = \dim_P^+ = \alpha_u^+$. For every nonzero spectral measure, Theorem C(2) gives the same conclusion under the additional condition $\theta + \alpha \in G_\alpha$.

In parts 2 and 3, $\theta = 1/4$ and the denominators of α are eventually odd. Then $1/4 \in G_\alpha$, as shown in the proof of part 2, and also $1/4 \in G_\alpha - \alpha$: since $q_n\alpha = p_n + \delta_n$, we have $|\sin(2\pi q_n(1/4+\alpha))| = |\cos(2\pi\delta_n)| \ge 1/2$ for large n. So, when $\beta(\alpha) < \infty$, Theorem C(2) shows that the exponent is also the packing dimension of every nonzero spectral measure of $H_{2,\alpha,1/4}$. When $\beta(\alpha) = \infty$, the exponent is 1, and so is every such packing dimension: Theorem C(1) gives $d^+ \ge 1$ almost everywhere, and every finite Borel measure on $\mathbb{R}$ has upper packing dimension at most 1, as noted after Theorem C. Hence in parts 2 and 3 the packing dimensions of all nonzero spectral measures of $H_{2,\alpha,1/4}$ take exactly the values stated for α_u^+. In particular the explicit operator of part 3 has spectral measures whose packing dimension has Turing degree $\mathbf{0}''$.

The noncomputability concerns infinite-time asymptotics of an infinite system. Finite-volume propagator entries and finite-time probabilities, at computable times and with specified computable data, can be approximated effectively to any prescribed accuracy. The theorem says that the long-time exponent cannot be extracted by any algorithm, even one with access to the halting problem.

11 Discussion and open problems

The dynamical picture. Damanik and Tcheremchantsev proved that the lower transport exponents vanish for every $\lambda > 1$ and every frequency [11]. Theorem A identifies the upper exponents. Rigorously: for $0 < \log\lambda < \beta < \infty$ and $\theta \in G_\alpha$, all lower transport exponents are 0, while every upper exponent $\beta^+(p)$, and the fastest one α_u^+, equals $1 - \log\lambda/\beta$. When $\beta \le \log\lambda$ the upper exponents are 0 as well. This is strong intermittency in the window $1 < \lambda < e^\beta$. Near the resonant times $T \approx |q_n\alpha - p_n|^{-1}$ of strongly Liouville scales the packet makes a burst. By Theorem B and Remark 9.8, a fixed positive fraction of the Abel-averaged packet moves out with speed about λ^{-q_n} and reaches distance about $T^{1-\log\lambda/b_n}$ at $T = |q_n\alpha - p_n|^{-1}$, up to factors $e^{\zeta q_n}$ with any fixed $\zeta > 0$, while the mass beyond $2e^{\zeta q_n}T^{1-\log\lambda/b_n}$ is exponentially small in q_n. We do not know how the rest of the packet is distributed. The upper exponents record the largest of these bursts. The vanishing lower exponents reflect times at which the packet is confined, as in Proposition 6.1. We prove nothing about the proportion of times spent in each regime.

Universality. The lower bound of Theorem 8.1 holds for every Lipschitz sampling function whose Lyapunov exponent is continuous near a spectral minimizer, with $\log\lambda$ replaced by the minimal Lyapunov exponent ℓ on the spectrum. The rest of this paper uses the almost Mathieu operator in an essential way: Chambers' formula makes the phase dependence of the discriminant a single cosine with an energy-independent amplitude, and both the upper bound and the arcsine law of Section 9 rest on the resulting conserved polynomial. A companion paper [12] proves that the fastest exponent equals $\max\{0, 1 - \ell_*/\beta\}$, with ℓ_* the minimal Lyapunov exponent on the spectrum, for every real-analytic potential that is supercritical and of type I on its spectrum. The frequency must be normalized: if $v(x) = w(kx)$ for an integer $k \ge 2$, the operator is the one with sampling function w and frequency $k\alpha$, and $\beta(k\alpha)$ can differ from $\beta(\alpha)$. The companion paper also reduces the law for every real-analytic potential with positive Lyapunov exponent to an explicit separation hypothesis on the periodic approximants, which remains open beyond type I.

Hausdorff versus packing dimension. Jitomirskaya and Last proved that for $\lambda > 1$ the spectral measures of the almost Mathieu operator have zero Hausdorff dimension [16]. By Theorem C, in the window $1 < \lambda < e^{\beta}$ their packing dimension is $1 - \log\lambda/\beta > 0$ for almost every phase. The two fractal dimensions thus split exactly as the lower and upper transport exponents do. Finer invariants of these measures, such as the generalized Rényi dimensions that bound transport from below [3], are not determined here.

Exceptional phases. The upper bounds are proved for $\theta \in G_\alpha$, and for $\theta \in G_\alpha \cap (G_\alpha - \alpha)$ in the case of packing dimensions. All lower bounds hold for every phase. At phases where $\sin(2\pi q_n\theta)$ is unusually small for infinitely many n, for example $\theta = 0$, where it vanishes for every n, the drifting cosine has a turning point near the origin and the barrier argument must be modified. Whether the exponents or the packing dimension can be larger at such phases is open.

The size of the fraction. The fraction $2^{-32}/(1+4\pi\lambda)$ in Theorem B comes from crude estimates. Its optimal value is open. So is the behaviour well after the resonant time: for $T \gg |q_n\alpha - p_n|^{-1}$ the elliptic runs are limited by the drift $|q_n\alpha - p_n|$ rather than by the damping $1/T$, and our guaranteed fraction degrades by the factor $T|q_n\alpha - p_n|$. This does not affect the exponents, which only need the resonant times.

Critical coupling. At $\lambda = 1$ the Lyapunov exponent vanishes. For $\beta(\alpha) > 0$, Jitomirskaya and Zhang proved quasiballistic transport at every phase [18, Corollary 3]. For $\beta(\alpha) = 0$ both halves of our method fail: the velocity λ^{-q} of the upper bound becomes 1, and the elliptic runs need scales with $|q_n\alpha - p_n| \le e^{-cq_n}$ for some $c > 0$, which then do not exist. Transport at critical coupling with $\beta(\alpha) = 0$ is a separate problem.

Physical meaning. Laboratory realizations of the Aubry–André model use frequencies such as the golden mean, for which $\beta = 0$ and Theorem A confirms the absence of power-law transport for $\lambda > 1$. The theorem shows that this absence is arithmetic, not a consequence of $\lambda > 1$ alone. A frequency with one exceptionally large partial quotient after a denominator q produces a transient burst of the kind described above over the time window $\lambda^q \lesssim T \lesssim |q\alpha - p|^{-1}$. Theorem B shows that, along strongly resonant scales, such bursts move a fixed fraction of the packet and not only a small tail. The exponent itself is an infinite-time quantity of an infinite system. Its noncomputability in Theorem D does not extend to finite-time probabilities at computable times, which can be approximated effectively to any prescribed accuracy.

How this work was done

The authors developed the results and proofs with assistance from AI models from more than one developer. They take responsibility for every claim. The authors chose the question, directed the work and decided which lines to pursue; the models did most of the exploration, derivation and drafting.

The proof was developed over several long working sessions, each building on the previous ones and recording its proofs, failed routes and self-checks. Each step was then checked by a separate session that had not seen the material, with instructions to refute it and to read the primary sources it relies on. The repairs found in these checks are incorporated here. Priority and literature searches were AI-assisted. No computation is used in any proof.

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